

Efficiency and Redistributive Effects of Progressive Housing Taxation*

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Abstract

Is it optimal to introduce progressivity in housing taxation? We explore this question by constructing a heterogeneous agent model featuring housing and entrepreneurship, calibrated to the Spanish economy. Our results indicate that a progressive housing tax can significantly enhance welfare and influence the economy through multiple channels. By imposing higher taxes on high-value properties, the policy curbs housing demand and lowers house prices—particularly benefiting lower-income and younger households. Moreover, it encourages rich households to reallocate savings from housing toward productive capital, thereby stimulating investment, output, and wages. In an economy with entrepreneurs the optimal tax design combines a flat income tax with a highly progressive housing tax, as elevated income tax rates at the top discourage business expansion. The equilibrium price effects and resulting welfare gains from this policy are amplified when housing supply is more inelastic.

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1 Introduction

Housing is the single most important asset for most households and the most important asset in the aggregate as it represents roughly 50% of national wealth in a typical economy (Jordà et al., 2019). Consequently, the housing stock represents a substantial potential tax base; yet, housing taxation typically contributes only modestly to public revenues (OECD, 2022). In this paper, we aim to explore the ramifications of housing taxation, with an emphasis on a progressive housing tax schedule. In particular, our goal is to explore whether a progressive housing tax schedule might be optimal and how this progressivity interacts with progressive income taxation, which is the main redistributive tax instrument in most economies.

To address this question, we develop a rich heterogeneous-agents, general equilibrium model with incomplete markets, incorporating both occupational and housing choices. The framework combines standard models of entrepreneurship (e.g., Cagetti and De Nardi (2006); Brüggemann (2021)) with heterogeneous-agent models that feature housing assets, such as Gervais (2002), Sommer et al. (2013), and Kaplan et al. (2020). Introducing progressive housing taxation into this setting generates several important mechanisms. First, housing taxation differentiates between workers and entrepreneurs with similar income and net worth, as workers typically hold a larger share of housing in their portfolios and thus face a higher tax burden. Second, when housing supply is inelastic, taxing high-value housing reduces aggregate demand and lowers house prices, which benefits poorer households. Third, the decline in house prices diminishes the collateral value available to entrepreneurs, thereby constraining their capacity to scale up business operations.

The model is calibrated to match a number of salient facts about the Spanish economy. Spain is an appropriate laboratory due to its large share of housing in the aggregate wealth: the median household allocates about 80% of its portfolio to housing wealth. The model generates realistic wealth and income distributions and broadly matches the portfolio share and homeownership rate across different dimensions, such as age, income and wealth. We introduce progressive housing taxation using a flexible functional form adapted from Heathcote, Storesletten, and Violante (2017). Employing this specification, we compute the optimal effective housing and income tax functions. All of our experiments are budget-neutral reforms. Welfare is measured by the consumption equivalent gains of the newborn household in expected utility and we only rely on steady state comparisons.

In our main experiments, we consider the case where subsidies are permitted in both housing and income tax schedules. In this setting, both optimal income and housing tax schedules exhibit strong progressivity and the reforms yield considerable welfare gains of 11.9% and 10%, respectively, in consumption-equivalent terms. However, these gains arise through distinct economic channels. A highly progressive income tax effectively redistributes income from high- to low-income households but leads to a substantial decline in the economy’s capital stock. This decline is primarily driven by reduced capital investment among the entrepreneurs, whose incentives to scale their businesses are weakened by high marginal tax rates. In contrast, the optimal housing tax schedule encourages households to reallocate resources away from housing, resulting in an increase in aggregate capital stock, with relatively stable demand for capital from self-employed. Moreover, the housing tax performs better in smoothing consumption over the life cycle, owing to the imperfect correlation between income and housing: individuals tend to earn more during working age but continue to reside in large homes during retirement. When subsidies are disallowed, housing taxation proves markedly superior to income taxation. The latter fails to benefit the poorest households, who already face no tax liabilities in the baseline economy. In contrast, optimal housing taxation without subsidies curbs housing demand among the wealthy, leading to significant declines in house prices—ultimately benefiting renters and young first-time buyers.

When jointly optimizing the income and housing tax schedules under a regime that allows subsidies, we find that the optimal policy entails a high, flat income tax rate of 59%, with revenues redistributed via a highly progressive housing tax. This combined approach balances the advantages and drawbacks of optimizing either tax in isolation. Crucially, the jointly optimal schedule preserves the capital stock close to its baseline level, with the decline in self-employed capital substantially smaller than under redistribution through the income tax alone. Although wages decline by 5%, house prices fall by 27%, substantially improving housing affordability. As a result, the homeownership rate rises by more than 18 percentage points. This policy configuration effectively smooths consumption both across the life cycle and within age groups, yielding welfare gains of 20%.

Our model features two key elements that prove important for determining the shape of the optimal tax schedule, both quantitatively and qualitatively: (i) the presence of entrepreneurs, and (ii) housing tenure choice. Entrepreneurship emerges as the most influential factor: in a counterfactual model without entrepreneurs, but allowing for tenure choice, the optimal policy reverses, favoring a flat housing tax and a highly progressive income tax. Moreover, in a model without tenure choice—in which all households are renters—but with entrepreneurs, it remains optimal to impose a flat income tax and a progressive housing tax. Nevertheless, we find that modeling homeownership matters for the results quantitatively: the stronger the preference for homeownership, or the larger the minimum housing size, the more progressive the optimal housing tax. Lastly, we show that lower housing supply elasticity amplifies both the optimal progressivity of housing taxation and the associated welfare gains. In essence, progressive income taxation imposes significant costs in the presence of entrepreneurs, as it dampens their incentives to expand their businesses. Conversely, progressive housing taxation lowers the opportunity cost for wealthy entrepreneurs of investing in their businesses, while simultaneously reducing house prices—benefiting poorer households, especially when land is scarce.

Related Literature. This paper contributes to four related strands of the optimal-taxation literature in heterogeneous-agent models. The first studies entrepreneurship and the optimal progressivity of the tax system. The second compares wealth taxation with capital-income taxation. The third analyzes the tax treatment of housing in Aiyagari–Bewley environments, including the design of housing taxes. The fourth examines consumption taxation, particularly recent work on progressive consumption taxes.

Entrepreneurship models are widely used to study the effects of tax changes, which is not surprising given that these models are able to match the empirical concentration of wealth and income, see [Quadrini \(2000\)](#) and [Cagetti and De Nardi \(2006\)](#). Our model more closely resembles the works of [Brüggemann \(2021\)](#) and [İmrohoroglu et al. \(2023\)](#). Both studies focus on the progressivity of income taxation introducing entrepreneurship and superstar shocks, which allow them to match the share of entrepreneurs at the top of the income distribution. The authors point out that the modeling of entrepreneurs is key in order to avoid overestimating the optimal progressivity. We contribute to this literature by noting that the modeling of entrepreneurs is not only crucial for income taxation, but also an important feature in the design of housing taxes. Crucially, we then show that this connection runs in both directions: just as entrepreneurship is key to understanding optimal housing taxes, the inclusion of housing is key to the optimal design of income taxation. When income and housing taxes are optimized jointly in our framework, the resulting policy differs markedly from prior findings: the optimal income tax is nearly flat, while the housing tax becomes the primary redistributive tool.

Since housing constitutes a form of wealth, our paper also relates to [Guvenen et al. \(2023\)](#) and [Boar and Midrigan \(2023\)](#), who study the relative merits of wealth versus capital income taxation. [Guvenen et al. \(2023\)](#) find that in an Overlapping Generations model economy with financially constrained entrepreneurs, shifting taxation from capital

income to wealth improves capital allocation and is therefore optimal. [Boar and Midrigan \(2023\)](#), by contrast, show that in an infinite-horizon model in which private businesses coexist with a corporate sector, the optimal policy reverses: taxing capital income becomes preferable, as the efficiency losses from misallocation are less severe. Our paper combines elements of both frameworks, building a life-cycle model with financially constrained entrepreneurs and a corporate sector. Unlike either study, our model incorporates housing and treats self-employment income as labor income for tax purposes; we find that the optimal policy combines a nearly flat income tax with a highly progressive housing tax.

We also build on the literature on the tax treatment of housing in Aiyagari–Bewley models. [Gervais \(2002\)](#) demonstrates that eliminating the preferential tax treatment of housing in the US—that is, removing mortgage interest deductibility and taxing imputed rents—while balancing the government budget through a lower income tax rate yields welfare gains in a model with fixed house prices. [Sommer and Sullivan \(2018\)](#), in a model with endogenous rental and house prices, find that eliminating mortgage interest deductibility would lower house prices, raise homeownership, and improve welfare. [Keane and Liu \(2025\)](#) incorporate housing investment and house price uncertainty into a life-cycle model and find that budget-neutral reforms—taxing imputed rents or eliminating mortgage interest deductibility—are welfare improving.

More closely related to our work, and focusing specifically on housing taxation, [Balke et al. \(2025\)](#) find that shifting from capital income taxation to a flat tax on housing in the US would generate welfare gains, primarily through capital deepening. The authors show that the reform largely benefits today’s young and low-income households while imposing one-time wealth losses on incumbent homeowners, and they study how it could be phased in to secure the support of a majority of households during the transition. We extend their analysis by examining budget-neutral reforms of progressive housing and income tax-and-transfer schedules. [Chiocchio \(2024\)](#) finds that the property tax in Italy is in fact regressive, owing to differences in assessment ratios across the housing distribution: the ratio of cadastral to market values is lower for higher-value homes. He builds a heterogeneous-agents model with housing to examine the welfare implications of various housing tax reforms. Among these, he studies progressive housing taxation, employing the same tax schedule for housing as for income, formulated as in [Heathcote et al. \(2017\)](#). This is the only other study that uses this type of progressive tax function to model housing taxation.¹ Although he does not conduct an optimal-taxation analysis, he establishes that increasing the progressivity of housing taxation pushes house prices down, raises the homeownership rate, and delivers net welfare gains. Moreover, the absence of occupational choice in his model neglects the important housing-as-collateral channel.

The closest studies to the present work are [Juan \(2018\)](#) and [Rotberg \(2022\)](#). To our knowledge, these are the only other Aiyagari–Bewley type models with housing and entrepreneurship. Both show that a wealth tax that taxes capital and housing at the same rate hurts capital allocation and welfare². Furthermore, [Rotberg \(2022\)](#) shows that allowing for a progressive wealth tax with high exemption thresholds and steep marginal rates at the top—while taxing housing and capital at different rates—reverses this result. The optimal policy taxes capital wealth at about 3% and housing at very high rates (around 288%) while exempting most households from both taxes. The high exemption thresholds allow poorer households to benefit from lower house prices and higher wages, while the differential taxation induces wealthy households to reshuffle their portfolios from housing toward capital. We complement these papers in several key directions. First, we analyze the joint design of housing and income taxation, which is typically the prime redistributive tool in most economies. Second, we employ a flexible functional form for both income and housing progressive tax

¹[Cho et al. \(2024\)](#) use the same tax function to estimate effective property transaction tax rates in Australia.

²In a different setting, [Borri and Reichlin \(2021\)](#) also conclude that a uniform wealth tax on financial and housing assets decreases welfare.

schedules, in the spirit of [Heathcote et al. \(2017\)](#), and we solve for the jointly optimal tax design. Crucially, we show that the result of a nearly flat income tax combined with a highly progressive housing tax hinges on the inclusion of entrepreneurs in the model.

In our model, housing delivers direct utility to households and therefore housing taxation functions as a form of consumption tax. Within the extensive literature on consumption taxation, the papers most closely related to ours are those that focus on optimal consumption taxes that incorporate progressivity. [Conesa, Li, and Li \(2020\)](#) analyze replacing income taxes with two flat consumption tax rates—one on basic goods and one on non-basic (luxury) goods—in a framework with endogenous labor supply and find that, despite a 10% increase in aggregate output, the policy reduces overall welfare. By contrast, again working in a model with endogenous labor supply, [da Costa and Santos \(2023\)](#) allow for progressive consumption taxation in the style of [Heathcote et al. \(2017\)](#), and report welfare gains of about 13% from replacing income taxes with a progressive consumption tax, a magnitude comparable to the welfare improvements we obtain under progressive housing taxation in this paper.

The remainder of the paper is organized as follows. Section 2 develops the model, and section 3 describes its parametrization. Section 4 describes how our model replicates the relevant patterns of the Spanish economy, section 5 presents the main counterfactual experiments, and section 6 examines the forces driving the results. Section 7 concludes.

2 Model

We develop a general-equilibrium life-cycle economy with incomplete markets, housing, and occupational choice. Households are finitely lived and differ in age, financial wealth, housing wealth, labor productivity, and entrepreneurial ability. Working-age households choose consumption, savings, housing services, and their occupation.

The model has two assets. Households can save in a financial asset and own housing. Housing provides dwelling services, is costly to adjust, and can be used as collateral for business activity. Households that do not own a home rent housing services from a rental intermediary. Entrepreneurs operate a decreasing-returns technology using capital and labor, and their scale is limited by a collateral constraint. The economy also includes a corporate sector, a construction sector, and a government that taxes income, consumption, inheritances, housing transactions, and housing.

2.1 Demographics and preferences

The economy is populated by a continuum of finitely lived households who age deterministically. Households survive from age j to $j + 1$ with probability ζ_j and at age T die with certainty. Utility comes from consumption c , dwelling services d and bequests b . The per-period utility flow at age j is given by a CRRA utility over a Cobb-Douglas composite of consumption and housing services:

$$u_j(c, d) = e_j \frac{\left(\frac{c^\xi d^{1-\xi}}{e_j}\right)^{1-\gamma}}{1-\gamma} \quad (1)$$

where ξ is the relative utility share of consumption in the utility of the household and γ represents the inverse of the intertemporal elasticity of substitution. Parameter e_j is an equivalence scale which adjusts for household size over the life cycle. Households also have a warm-glow bequest motive on the net wealth b as in [De Nardi \(2004\)](#):

$$\Phi(b) = \phi_1 \frac{(\phi_2 + b)^{1-\gamma}}{1-\gamma}$$

The parameters ϕ_1 and ϕ_2 capture bequest intensity and the luxury nature of bequests, respectively. We do not model intergenerational links. Instead all the bequests are aggregated and then divided among the remaining households based on their age and labor ability. This allows us to have a realistic wealth distribution among retirees and model intergenerational transfers without tracking an additional state for the amount of inheritances received by the newborns ($j = 1$). Without the bequest motive households would tend to deplete their wealth, while this is not supported by the data.

2.2 Labor productivity and entrepreneurial ability

Working-age households face idiosyncratic and persistent labor productivity (z) and entrepreneurial ability (θ) shocks, realized at the beginning of every period. The two shocks are independent and evolve over time according to a first-order Markov process with transition probabilities $\Lambda_z(z'|z)$ and $\Lambda_\theta(\theta'|\theta)$, respectively.

Total labor productivity also includes an age-dependent deterministic component $q(j)$. Upon retirement at age R , the household receives a fixed pension that is a share $\bar{p}$ of the labor income received just before retirement.

$$n_j(z) = \begin{cases} q(j) z & \text{if } j < R \\ \bar{p} q(R-1) \bar{z} & \text{if } j \geq R \end{cases} \quad (2)$$

where $\bar{z}$ denotes the realization of the labor productivity shock upon retirement.

2.3 Tax system

We model the main features of the Spanish fiscal system. Labor and self-employment income are taxed progressively, with a zero-tax region for low earners. Following [Heathcote et al. \(2017\)](#) we model the tax payment as:

$$t_y(y) = \max\{0, (1 - \tau_y^l (\frac{y}{\bar{y}})^{-\tau_y^p}) y\} \quad (3)$$

where $y/\bar{y}$ is the ratio between the household's labor/self-employment income after paying the social security contributions and the average income of the economy. The kink in the function reflects the fact that a substantial share of the population does not pay any income taxes. Notice that the two parameters τ_y^l and τ_y^p govern the level and the progressivity of the progressive income tax function, respectively. If there were no progressivity ($\tau_y^p = 0$) in the tax, then the income tax rate would correspond to the $1 - \tau_y^l$ parameter.

Working-age households also have to pay payroll taxes up to a cap. As in [Fuster \(2022\)](#), we model the social-security contributions as:

$$t_{ss}(y) = \min\{\tau_{ss} y, \tau_{ss} \hat{I}_{ss}\} \quad (4)$$

where $\hat{I}_{ss}$ is the maximum payroll tax base.

We choose to model capital income as a flat tax rate τ_k . This approximation is motivated by evidence that, despite

the formally progressive taxation of savings income in Spain, effective tax rates display only limited progressivity (Carrillo & Ramos, 2024).

In Spain, inheritances are taxed progressively. To reflect this fact we model the total amount of inheritance taxes as a stepwise function:

$$t_{\text{inh}}(b) = \begin{cases} \tau_{\text{inh}}^1 b, & \text{if } 0 (\equiv B_0) \leq b \leq B_1, \\ \sum_{j=1}^{k-1} \tau_{\text{inh}}^j (B_j - B_{j-1}) + \tau_{\text{inh}}^k (b - B_{k-1}), & \text{if } B_{k-1} < b \leq B_k, \quad k = 2, \dots, n-1, \\ \sum_{j=1}^{n-1} \tau_{\text{inh}}^j (B_j - B_{j-1}) + \tau_{\text{inh}}^n (b - B_{n-1}), & \text{if } b > B_{n-1}, \end{cases} \quad (5)$$

Finally, housing is subject to two sets of taxes. First, there is a transaction tax. This tax is paid on the value of the purchased home and we set it to be a flat tax rate τ_h^t . Second, we introduce a progressive housing tax employing the same functional form as the income tax, adapted from Heathcote et al. (2017). In particular we choose the tax burden to be $t_h(h)p_h$, with:

$$t_h(h) = (1 - \tau_h^l (\frac{h}{\bar{h}})^{-\tau_h^p}) h \quad (6)$$

where as before the parameter τ_h^l governs the tax level and τ_h^p the progressivity of the housing tax schedule.³ The parameter $\bar{h}$ represents the mean housing size in the baseline economy. When $\tau_h^p = 0$, the recurrent housing tax collapses to a flat rate taxation, which is the case in the baseline economy. When τ_h^p is positive, all households with houses larger than $\bar{h} \tau_h^{\frac{1}{\tau_h^p}}$ are paying the tax, while the rest are receiving subsidies.⁴ For a rented property the housing tax (or subsidy) is paid (received) by the renter.

Households also pay a flat consumption tax τ_c .

2.4 Entrepreneurs

An entrepreneur with entrepreneurial ability θ , labor productivity z and using k units of capital and l labor units (in efficiency terms) produces an amount of output equal to:

$$y_j^e(z, \theta, k, l) = \theta (k^\alpha (n_j(z) + l)^{1-\alpha})^\nu$$

where $\alpha \in (0, 1)$ is the share of capital and $\nu < 1$ is the span-of-control parameter which governs the degree of returns to scale. Notice that we assume that the entrepreneur supplies all of her own effective labor to the firm. The total net-of-taxes income of an entrepreneur is given by:

³Since all housing units trade at the same price p_h , the tax rates expressed in terms of housing size are equivalent to those expressed in terms of housing value: $\frac{p_h t_h(h)}{p_h h} = (1 - \tau_h^l (\frac{h}{\bar{h}})^{-\tau_h^p})$.

⁴In the appendix, we also present results for which we don't allow for subsidies.

$$\begin{aligned}
\Pi_j^e(a, h, z, \theta) &= \max_{k, l} \{ \pi_j^e - t_{ss}(\pi_j^e) - t_y(\pi_j^e - t_{ss}(\pi_j^e)) + (a - k)r(1 - \tau_k)\mathbb{1}_{k < a} \} \\
\text{s.t. } \pi_j^e &= y_j^e(z, \theta, k, l) - \delta_k k - (r + \iota)(k - a)\mathbb{1}_{k \geq a} + m - wl \\
m &= -a(r + \iota)\mathbb{1}_{a < 0} \\
0 &\leq k \leq a + \lambda(p_h h + a) \\
l &\geq 0
\end{aligned} \tag{7}$$

where π^e stands for the gross business profits. The total income for entrepreneurial households is given by their net-of-taxes profits plus the net income from savings. Gross profit is just the total production of the entrepreneur, minus capital depreciation ($\delta_k k$), labor costs (wl) and the cost of financing capital for constrained entrepreneurs. The term m is just a slight adjustment so that the mortgage costs are not imputed into the business profits. Entrepreneurs will earn capital income only if the capital used in production does not exceed the amount of financial assets that the household has.

The amount of capital used by entrepreneurs is limited by the collateral constraint. The maximum capital is given by the sum of their financial assets and a fraction of their total wealth. The main difference with respect to a one-asset model is that here entrepreneurs can transform one unit of financial asset into capital, but they cannot do the same with housing. Nevertheless, entrepreneurs can use housing as collateral to rent additional capital. Hence, the housing value of an entrepreneur may influence their optimal capital decisions.⁵

Borrowing is costly and entails a spread ι over the interest rate r , reflecting a wasteful intermediation cost.

2.5 Dynamic program

A household of age j enters the period with financial assets a and housing assets h carried over from the previous period. At the beginning of the period, it observes two idiosyncratic shocks: a labor productivity shock z and an entrepreneurial ability shock θ . The household's problem is thus defined over five individual state variables: (a, h, z, θ, j) . If the household received inheritance it does not enter its budget constraint, instead it affects only the amount of financial assets that the household has. Thus, one can think about the financial assets represented here, a , as inheritance-adjusted. To simplify the decision problem, we divide it into two sequential stages. In the first stage, the household makes consumption, savings, and occupational choices. In the second stage, it decides on whether to adjust its housing status for the next period.

In what follows, we present the two-stage problem for working-age ($j < R$) households. The problem of retired households is analogous, with three exceptions: entrepreneurship is not available to retirees, retirees cannot incur debt and labor productivity is fixed. This problem is reported in the Appendix B.1.

First stage: the household decides on current consumption c , financial savings to carry into the next stage s , its occupational status $i \in \{w, e\}$ (worker or entrepreneur), and rental housing services h^r . The rental choice is only

⁵There is plenty of empirical evidence supporting this channel. For the Spanish case Basco et al. (2025) note that there is positive correlation between the growth of housing price and the growth of credit and investment.

relevant when the household is currently a renter ($h = 0$).

$$\mathcal{V}_j(a, h, z, \theta) = \max_{c, s, h^r, i} u_j(c, d) + \tilde{\mathcal{V}}_j(s, h, z, \theta)$$

$$\text{s.t. } c(1 + \tau_c) + s + (p_r h^r + t_h(h^r)p_h)\mathbb{1}_{h=0} + m \leq a + y_j^i - p_h h \delta_h - p_h t_h(h) \quad (8)$$

$$m = -(r + \iota)a\mathbb{1}_{a < 0} \quad (9)$$

$$i \in \{w, e\} \quad (10)$$

$$y_j^i = \begin{cases} wn_j(z) + r(1 - \tau_k)a\mathbb{1}_{a \geq 0} - t_y(wn(j, z) - t_{ss}(wn(j, z))) - t_{ss}(wn(j, z)) & \text{if } i = w \\ \Pi_j^e(a, h, z, \theta) & \text{if } i = e \end{cases} \quad (11)$$

$$d = h^r \mathbb{1}_{h=0} + \chi h \mathbb{1}_{h > 0} \quad (12)$$

$$s \geq \min\{a, 0\} \quad (13)$$

$$h^r \geq 0 \quad (14)$$

Households derive utility from non-durable consumption (c) and housing services (d). As defined in equation (1), utility is adjusted by an age-dependent equivalence scale factor e_j , to reflect the fact that household size changes along the life-cycle. The continuation value $\tilde{\mathcal{V}}_j$ is the result of the second stage described below.

Dwelling services are produced linearly from either rented (h^r) or owned (h) housing units, see equation (12). Renters obtain one unit of housing service per rented unit, while owners receive χ units per unit of housing, introducing a preference or quality premium for ownership.

The budget constraint of working-age households is given by equation (8). The right-hand side represents the resources of the household, which is the sum of financial assets (a) and disposable income. Notice that $\Pi_j^e(a, h, z, \theta)$ is the net income of entrepreneurial households that comes from the problem (7). The other case in equation (11) represents the worker's net income. This income is the gross wage $wn_j(z)$ and net capital income minus progressive income taxes and social security contributions. Occupational choice is endogenous in our model. Nevertheless, since there are neither entry nor exit costs the occupational decision becomes a static one: the household compares the net income from working (after labor income and payroll taxes) with expected profits from entrepreneurship $\Pi_j^e(a, h, z, \theta)$. It chooses to be an entrepreneur if the latter exceeds the former.

Regardless of their occupational status, homeowners incur housing maintenance and property tax expenses, where maintenance is given by the housing depreciation rate δ_h . In the case of rented housing, renters pay the housing tax (or receive the housing subsidy), calculated as a fraction of the value of the dwelling they occupy, analogous to the treatment for homeowners. As in the entrepreneurs' problem, indebted households pay interest at the deposit rate augmented by the wasteful intermediation cost ι . Households also pay a consumption tax ($\tau_c c$).

Second stage: the household decides whether to adjust its housing holdings for the next period. The objective of the household is to maximize the expected value function for the next period, taking into account the distribution of idiosyncratic shocks and the survival probability ζ_j . The next-period value function is discounted using the subjective discount factor β .

The resources available for the household are the savings carried from the first stage s and the value of their current

housing, which comprises the right-hand side of the budget constraint in equation (15).

$$\tilde{\mathcal{V}}_j(s, h, z, \theta) = \max_{h', a'} \beta \zeta_j \mathbb{E}_{z', \theta'} [\mathcal{V}_{j+1}(a' + B_{j+1}(z'), h', z', \theta')] + \beta(1 - \zeta_j) \Phi(b)$$

$$\text{s.t. } j < R$$

$$p_h h' + p_h C_{adj}(h, h') + a' \leq s + p_h h \quad (15)$$

$$C_{adj}(h, h') = \mathbb{1}_{h \neq h'} (\psi_s h + (\psi_b + \tau_h^t) h') \quad (16)$$

$$b = p_h h' + a' - \psi_s p_h h' - t_{inh}(p_h h' + a') \quad (17)$$

$$a' \geq -\psi p_h h' \quad \text{if } h \neq h' \quad (18)$$

$$h' \in \hat{\mathcal{H}} = \{0, h_1, h_2, \dots, h_N\} \quad (19)$$

Notice that the household is subject to several restrictions. First, there are transaction costs. As stated in equation (16), a household that chooses to change their current dwelling has to pay transaction costs for buying ψ_b and selling ψ_s property. Additionally, the household has to pay a transaction tax, τ_h^t , which is effectively a buying cost for the household. This type of non-convex transaction cost is standard in the literature, see [Sommer et al. \(2013\)](#) or [Cho et al. \(2024\)](#). Second, there is a loan-to-value (LTV) restriction for mortgages. A household can borrow up to a ψ fraction of the value of the house. This simple constraint allows us to model access to credit markets to purchase a house without including the whole mortgage repayment system to keep the state space limited.

Third, to reflect the fact that housing is a lumpy investment, h' is restricted to be an element of a grid $\hat{\mathcal{H}}$. If a household chooses $h' = 0$, then the next period it becomes a renter and chooses the amount of rented units in the first stage, as described above. In order to become a homeowner the household has to buy at least h_{min} housing units.

As for bequests (b), notice that upon death the household transforms the housing assets, if any, into financial ones and pays the associated selling costs as well as the nonlinear inheritances tax. In case the household survives, then it receives bequests $B_{j+1}(z')$ that depend on tomorrow's age and labor productivity.

2.6 Production

Corporate sector. The technology in the corporate sector is given by a standard Cobb-Douglas technology $Y_c = K_c^\alpha L_c^{1-\alpha}$. The representative corporate firm solves the following problem:

$$\max_{K_c, L_c} Y_c(K_c, L_c) - (r + \delta_k) K_c - w L_c \quad (20)$$

Construction firm. The construction firm uses capital, labor and land in order to produce housing units:

$$\Pi_h = \max_{K_h, L_h} \{p_h K_h^{\alpha(1-\gamma_L)} L_h^{(1-\alpha)(1-\gamma_L)} (\delta_h \bar{L})^{\gamma_L} - (r + \delta_k) K_h - w L_h\}$$

Where $\bar{L}$ is the total land in the economy. Thus, effectively each period a δ_h fraction of houses are destroyed and the same proportion of land becomes available.⁶ The share of land in the production of housing is denoted γ_L . Notice that

⁶Notice that we normalize construction firm's TFP to 1, and since we will choose to also normalize house prices $p_h = 1$ when calibrating, land in the economy $\bar{L}$ will be used for scaling to ensure clearing of the housing market. Alternatively we could have normalized $\bar{L}$ to 1, and

we assume that the relative share of capital and labor is the same as in the other sectors. After taking into account the first-order conditions this problem yields the following housing supply function:

$$I_h = p_h^{\frac{(1-\gamma_L)}{\gamma_L}} \left(\frac{\alpha}{r + \delta_k} \right)^{\frac{\alpha(1-\gamma_L)}{\gamma_L}} \left(\frac{(1-\alpha)}{w} \right)^{\frac{(1-\alpha)(1-\gamma_L)}{\gamma_L}} (1-\gamma_L)^{\frac{1-\gamma_L}{\gamma_L}} \delta_h \bar{L} \quad (21)$$

Since land is a fixed factor of production the construction firm has positive profits in equilibrium. As in [Favilukis et al. \(2017\)](#) and [Kaplan et al. \(2020\)](#), we assume that these profits are collected by the government. This effectively assumes that the government is the owner of all land and hence collects the rents from this fixed input, as if these lands/permits are sold to the construction firm in a competitive market.

Financial Intermediary. As in [Balke et al. \(2025\)](#), there is a financial intermediary that lives for only two periods and discounts profits at the interest rate r . At the end of period t , the households deposit their savings $\mathcal{D}'$ to the newly born intermediary and receive interest at rate r next period. The deposits are used to lend mortgages to households ($\mathcal{M}'$) and capital to firms and self-employed ($\mathcal{K}'$), as well as to purchase the rental stock ($p_h \mathcal{H}'_r$) from the dying financial intermediary. At the beginning of next period, the intermediary receives interest from borrowers, firms and self-employed, rental payments from renters and liquidates its housing stock.

For operating the rental market, the intermediary faces a zero cost of transforming the rental stock into apartments of different sizes, and rents these apartments to households at price p_r , while paying depreciation costs. It also faces an intermediation cost η , proportional to the size of the rental stock. The problem of the intermediary can be described as:

$$\begin{aligned} V_f = \max_{\mathcal{K}'_f, \mathcal{H}'_{f,r}, \mathcal{M}'_f, \mathcal{D}'_f} & \left\{ -p_h \mathcal{H}'_{f,r} - \mathcal{K}'_f - \mathcal{M}'_f \right. \\ & + \frac{(p_r \mathcal{H}'_{f,r} - \delta_h p_h \mathcal{H}'_{f,r} - \eta \mathcal{H}'_{f,r})}{1+r} \\ & \left. + \frac{p_h \mathcal{H}'_{f,r}}{1+r} + \frac{(1+r)\mathcal{K}'_f}{1+r} + \frac{(1+r)\mathcal{M}'_f}{1+r} \right\} \\ \text{s.t. } & \mathcal{K}'_f + p_h \mathcal{H}'_{f,r} + \mathcal{M}'_f = \mathcal{D}'_f, \end{aligned} \quad (22)$$

At equilibrium, the no-arbitrage condition between investing in the rental stock and investing in mortgages or capital implies that:

$$p_r = (\delta_h + r)p_h + \eta \quad (23)$$

and the financial intermediary has zero profits.

2.7 Government

The government collects the aforementioned set of taxes – as described in section 2.3 – from the households, denoted here as $T_{hh}(\cdot)$ for the sake of simplicity. This includes labor and capital income, payroll contributions, consumption, inheritances, and housing taxes.

use the construction firm's TFP as a scaling parameter, or we could have normalized both construction firm's TFP and $\bar{L}$ to 1, and allow house prices to adjust at the baseline calibration. The model's results are invariant to these normalization choices.

$$G + \int wn(j, z) \mathbb{1}_{j \geq R} d\Gamma(\mathbf{x}) = \int T_{hh}(\mathbf{x}) d\Gamma(\mathbf{x}) + \Pi_h \quad (24)$$

Notice that $\mathbf{x}$ denotes the state variables of the economy and Γ the stationary distribution of households, see the details below. Apart from the taxes levied on households, the government collects the profits of the construction firm (Π^h). These funds are used to fund a wasteful government expenditure G as well as the pension system.

2.8 Equilibrium definition

We focus on stationary competitive equilibrium. Let $\mathbf{x} = (j, a, h, z, \theta)$ be the individual's state vector, where j is the age of the household, a and h represent the current holdings of the financial and housing assets respectively, and z and θ stand for the current realizations of the labor productivity and entrepreneurial ability shocks.

An equilibrium is given by prices $\{r, w, p_h, p_r\}$ and decision rules for non-durable consumption $c(\mathbf{x})$; consumption of rental units $h^r(\mathbf{x})$; interim savings $s(\mathbf{x})$, occupational choices for working-age households ($j < R$) $\mathbb{1}_e(\mathbf{x})$, $\mathbb{1}_w(\mathbf{x})$; capital investment $k(\mathbf{x})$; efficient labor hired by entrepreneurs $l(\mathbf{x})$; housing assets $h'(\mathbf{x})$, and savings $a'(\mathbf{x})$, along with distribution of households over the state variables: Γ and a (received) bequests function $B_j(z)$, such that:

- Given prices and government tax schedules the policy functions $c(\mathbf{x})$, $h^r(\mathbf{x})$, $s(\mathbf{x})$, $\mathbb{1}_e(\mathbf{x})$, $\mathbb{1}_w(\mathbf{x})$, $k(\mathbf{x})$, $l(\mathbf{x})$, $h'(\mathbf{x})$ and $a'(\mathbf{x})$ solve the households maximization problems described above.
- Labor market clears: total efficient labor hired by the entrepreneurial sector $\int l(\mathbf{x}) \mathbb{1}_e(\mathbf{x}) d\Gamma(\mathbf{x})$, the efficient labor employed in the corporate sector L_c and efficient labor units hired by the construction firm L_h equal the total labor supplied by workers

$$\int l(\mathbf{x}) \mathbb{1}_e(\mathbf{x}) d\Gamma(\mathbf{x}) + L_c + L_h = \int \mathbb{1}_w(\mathbf{x}) n(j, z) d\Gamma(\mathbf{x}) \quad (25)$$

where $\mathbb{1}_w(\mathbf{x}) = 1$ if the agent is a worker and $\mathbb{1}_e(\mathbf{x}) = 1$ if the agent is an entrepreneur, and zero otherwise.

- The market for deposits clears: total savings in the financial asset $\int a(\mathbf{x}) d\Gamma(\mathbf{x})$ are either used as capital or borrowed by the rental intermediary to finance the purchase of the rental units.

$$K_c + K_h + \int k(\mathbf{x}) \mathbb{1}_e(\mathbf{x}) d\Gamma(\mathbf{x}) + p_h \int h^r(\mathbf{x}) d\Gamma(\mathbf{x}) = \int a(\mathbf{x}) d\Gamma(\mathbf{x}) \quad (26)$$

- Housing market clears: total demand of new housing units equals the housing supply I_h , defined by Equation (21).

$$\delta_h \int h^r(\mathbf{x}) + h(\mathbf{x}) d\Gamma(\mathbf{x}) = I_h \quad (27)$$

- The government budget – defined by Equation (24)– is satisfied.
- The total amount of bequests received equals the total aggregate bundle of bequests left:

$$\int B_j(z) d\Gamma(\mathbf{x}) = \int (1 - \zeta_j) \left[a'(\mathbf{x}) + p_h h'(\mathbf{x}) - \psi_s p_h h'(\mathbf{x}) - t_{\text{inh}}(p_h h'(\mathbf{x}) + a'(\mathbf{x})) \right] d\Gamma(\mathbf{x}). \quad (28)$$

- The distribution of households $\Gamma(\mathbf{x})$ is invariant. That is, it reproduces itself according to a given transition

function.

3 Parametrization

In this section, we explain how we map the model’s initial steady state to the data. First, Table 1 reports the parameters that we fix exogenously. When possible, we directly use the data to set their values. Our main data source is the Spanish Survey of Household Finances or EFF (*Encuesta Financiera de las Familias*). This is a detailed survey that reports detailed income and wealth information for Spanish households. We rely on the 2022 wave for most of our own calculations. Otherwise we rely on reported values in previous research. The remaining set of parameters are calibrated internally to match relevant moments from the data. Table 2 reports these parameters, while in Table 3 we report the value of the model-generated moments and their empirical counterparts. Now we discuss these parameters in more depth.

Demographics. The model period is one year. Households enter the model at age 20 and retire exogenously when they reach 65 years. At age 85 the household dies with certainty. Along their lifetime, households face an exogenous and known age-dependent survival probability ζ_j . This probability is taken from death rates published by the INE (the Spanish National Institute of Statistics). To make the data smoother we fit a fourth-degree polynomial on age. The results, along with all other regression results, are in Appendix D.1.

Preferences. Utility is adjusted by life-cycle changes in the household size. To obtain an age-dependent equivalence scale factor e_j we fit a standard OECD equivalence scale onto a third-order polynomial using the data from the EFF. The relative risk aversion parameter γ is set to 1.5, a standard value in the literature.

This leaves us with four preference parameters to calibrate internally. The discount factor β is calibrated to match the wealth-to-income ratio of the Spanish economy. The value of this moment is taken from Blanco et al. (2021), and we use the value corresponding to 2022. The two bequest parameters $-\phi_1$ and ϕ_2 are set to match the wealth distribution of retirees (measured by the Gini index) and the homeownership rate of households older than 65.

The relative preference for consumption goods over housing services ξ will help us to match the housing related moments in Table 3. In particular we use this parameter to target the average housing portfolio share. Notice that in the limiting case where ξ tends to one, agents do not value housing services and hence they do not accumulate any housing assets. Thus, this parameter heavily influences the housing portfolio share as well as the share of renters in the economy.

Inheritances. Our households have an explicit warm-glow preference for leaving bequests. When a household dies with a strictly positive net wealth, all those assets are bundled into an aggregate bundle $\mathcal{B}$. This aggregate inheritance is then redistributed across the households along two dimensions: age and labor productivity shock. We compute the share of inheritances by age using the EFF data, for estimation results see section D.1. The resulting double-humped function has two maxima: one at age 20 and another at 55. This reflects the fact that most households receive significant inheritances (such as a house) in their 50s. To reflect a degree of intergenerational persistence, we correlate the received inheritances with the labor ability of the household. We do so by scaling the value of inheritances received for each realization of the z shock, as the realization divided by the mean of the ergodic distribution.

Labor productivity and entrepreneurial ability. The deterministic component of labor productivity is estimated using the EFF data. We regress the logarithm of labor income on second order age polynomial (see Appendix D.1). As for the stochastic part, we assume that z can take 6 values. The first five are associated with normal labor earnings,

Table 1: Externally calibrated parameters

		Value	Source
Demographics and preferences			
γ	Relative risk aversion	1.5	Standard Value
ζ_j	Survival probabilities	-	INE
e_j	Equivalence scale	-	EFF
Labor productivity			
$q(j)$	Income deterministic component	-	EFF
ρ_z	Persistence of income shock	0.96	Standard Value
$\bar{p}$	Ratio pension to last income	0.75	EFF
Taxes			
τ_y^l	Income tax level	0.8817	(Carrillo & Ramos, 2024)
τ_y^p	Income tax progressivity	0.0996	(Carrillo & Ramos, 2024)
τ_c	Consumption tax rate	15%	Agencia Tributaria
τ_{ss}	Payroll contribution tax rate	28%	(Fuster, 2022)
$\hat{I}_{ss}$	Payroll maximum tax base	1.28 $\bar{y}$	Agencia Tributaria
τ_k	Capital income tax	20%	See text
τ_h^l	Housing tax level	0.9938	(Barrios Cobos et al., 2019)
τ_h^p	Housing tax progressivity	0	See text
τ_h^t	Housing transaction tax	8%	See text
Production			
ν	Span-of-control	0.759	(Fella et al., 2022)
λ	Collateral constraint	0.5	(Fuster, 2022)
α	Capital share	0.33	Standard Value
γ_L	Land share housing sector	0.46	See text
δ_k	Capital depreciation rate	0.06	Standard Value
Housing			
ψ	Maximum mortgage LTV ratio	0.8	Standard Value
ψ_s	Cost of selling a house	0.045	Global Property Guide
ψ_b	Cost of buying a house	0.029	Global Property Guide
ϵ_h	Supply price elasticity	1.17	(Cavalleri et al., 2019)
δ_h	Housing depreciation rate	0.015	Standard Value
ι	Financial spread	0.016	Banco de España

while z_6 represents a superstar shock, as in [İmrohoroglu et al. \(2023\)](#) and [Brüggemann \(2021\)](#). The first five grid points are assumed to follow a standard AR(1) process – in logs – with persistence ρ_z and standard deviation σ_z . We fix the persistence parameter to be 0.96, a standard value in the literature; while the dispersion parameter is used to match the Gini of the workers’ earnings distribution⁷.

This leaves us with the three superstar-shock parameters ($z_6, \pi_6^{z^+}, \pi_6^{z^-}$) which correspond to the value of the shock, the probability of reaching it from any of the first 5 grid points, and the probability of falling back to the medium ability level, z_3 . Since our model already features entrepreneurship, the superstar shock is not introduced to match the wealth distribution but to have the right distribution of entrepreneurs and workers at the top. Without this high level of labor ability the model would generate the counterfactual implication that almost 100% of households at the top 1% of the income distribution are entrepreneurs while this fraction is 29.6% in the Spanish data. Thus, we use these

⁷We could rely on empirical estimates for the standard deviation of workers earnings. Nevertheless, because the occupational choice is endogenous it could be the case that households with low z are underrepresented in entrepreneurship, thus affecting the effective earnings distribution among workers.

three parameters to match the share of entrepreneurs at the top 1% of the income distribution as well as the income shares of the top 1% and top 10%. The full transition matrix and the values of the z shock are reported in Appendix D.2. Notice that the value of our superstar shock, z_6 , is only 2.4 times the highest value of the AR(1) shock (z_5). As a result, our model does not rely on the superstar shock to replicate the empirical levels of wealth inequality.⁸

For the structure of entrepreneurial ability, we closely follow Brüggemann (2021). There are only four potential entrepreneurial abilities which are characterized by the parameters $\bar{\theta}$ and $\hat{\theta}$ such that $\theta = \{0, \bar{\theta}(1 - \hat{\theta}), \bar{\theta}, \bar{\theta}(1 + \hat{\theta})\}$. For the transition matrix we assume that a household can only transition into neighboring ability states:

$$\Lambda = \begin{pmatrix} \pi_1^\theta & 1 - \pi_1^\theta & 0 & 0 \\ \pi_2^\theta & \pi_3^\theta & 1 - \pi_2^\theta - \pi_3^\theta & 0 \\ 0 & \pi_2^\theta & \pi_3^\theta & 1 - \pi_2^\theta - \pi_3^\theta \\ 0 & 0 & \pi_4^\theta & 1 - \pi_4^\theta \end{pmatrix}$$

The six parameters characterizing the entrepreneurial ability process are calibrated internally to match 6 empirical moments: the fraction of entrepreneurs in the Spanish economy (7.73 percent); the Gini coefficient of entrepreneurs' income of 49.1; the share of employers among entrepreneurs (55.22 percent); the median income of entrepreneurs relative to workers (1.33); the share of entrepreneurs at the top 10% of the wealth distribution, and the Gini coefficient of net wealth of 68.50.

Taxation. The parameters of progressive income taxation are taken from the parametric estimates of Carrillo and Ramos (2024). The authors also provide an estimated kink function for the capital income tax. Nevertheless, the function is barely progressive and stabilizes at a 20% effective tax rate. Thus, we set $\tau_k = 0.2$.

The parameters for the Social Security contributions come directly from the Spanish Tax Agency (*Agencia Tributaria*). We set the tax rate to 28%, which corresponds to the statutory tax rate. The maximum tax base in 2022 was 4,139 euros monthly, which yearly translates into 1.28 times the average earnings. The same source reports that the effective consumption tax in Spain was roughly 15% in 2022.

We rely on Fuster (2022) to set the inheritance taxes: Table D2 in Appendix D.2 reproduces the statutory tax schedule of inheritances. We adjust the statutory tax rates to take into account different regional tax credits. On average, these credits are about 58% of the quota implied by the statutory tax.

The tax rate for housing transaction, τ_h^t , ranges from 6 to 10% depending on whether the housing unit is brand new or not. In our model there is no difference between new and old houses, the only important feature is the size. This is why we decide to set the value of $\tau_h^t = 8\%$, to be in the middle ground.

As for the effective property tax, we rely on the estimates of Barrios Cobos et al. (2019), who construct homogeneous estimates of the effective tax rates across European nations. In the baseline the housing tax is flat, implying that there is zero progressivity $\tau_h^p = 0$. Thus, the effective tax rate is just $1 - \tau_h^l$.⁹

These tax values in equilibrium imply a unique government spending value G that balances the budget. We solve for this value and keep this wasteful government expenditure fixed throughout the rest of our experiments.

⁸Our estimate is lower than, though closer to, the 5.5 times reported by İmrohoroglu et al. (2023), who also feature entrepreneurship, and substantially below the values exceeding 100 estimated by Castaneda et al. (2003) and Kindermann and Krueger (2022), who rely exclusively on superstar workers to match wealth concentration at the top.

⁹Although in most countries the housing taxes are flat at the local level it is difficult to determine whether this taxation is progressive or regressive due to several issues. The two most important ones are the variation of the taxes across municipalities and the fact that households are taxed on the cadastral value of the properties rather than the market one (for example see McMillen and Singh (2020)).

Table 2: Internally calibrated parameters

		Value
Housing		
χ	Homeownership utility premium	1.57
h_{min}	Minimum housing size for homeowners	71
Preferences		
ϕ_1	Bequest intensity parameter	266.3
ϕ_2	Luxury good parameter of bequests	1426
β	Discount factor	0.9705
ξ	Consumption utility share	0.6509
Entrepreneurial ability process		
$\hat{\theta}$	Dispersion of ent. ability	0.3501
$\bar{\theta}$	Average of ent. ability	4.341
$(\pi_1, \pi_2, \pi_3, \pi_4)$	Transition probabilities	(0.9441, 0.4734, 0.4786, 0.9889)
Labor productivity process		
σ_z	St. dev. labor productivity	0.6981
z_6	Highest labor prod. level	2.42 z_5
π_6^{z+}	Probability of reaching z_6	0.0021
π_6^{z-}	Probability of leaving z_6	0.1989

Technology. The capital share in the corporate and entrepreneurial sector is set to 0.33 and the depreciation rate in both sectors is fixed to $\delta_k = 0.06$, both being standard values in the literature. For the collateral constraint we set $\lambda = 0.5$. This implies that entrepreneurs can borrow an additional 50% of their net wealth to expand their business. Notice that not only is this a standard value in entrepreneurial models, but it also aligns with the empirical estimates of [García-Posada and Mora-Sanguinetti \(2014\)](#) for the small Spanish firms. The span-of-control parameter is set to 0.759 following the estimate of returns to scale in Spain by [Fella et al. \(2022\)](#).

Housing. The maximum loan-to-value for mortgages is set to 80%, standard value in the literature. For the housing depreciation parameter we set a value of 1.5%, which is standard in the literature ([Kaplan et al., 2020](#)). The mortgage debt spread ι is taken from the interest rates data provided by Banco de España (Spanish Central Bank). We estimate this spread as the difference between the average interest rate for new mortgages and the return of one to two year deposits. The cost of selling and buying property is taken from the Global Property Guide website, which provides a comprehensive overview of real estate markets on a country by country basis. The reported cost for selling a house is 4.5% and the one for purchasing a house is 10.9%. Nevertheless we fix $\psi_b = 0.029$ to reflect the fact that most of that cost is the transaction tax rate, which we have set to 8%.

The weight of land in the housing production function is pinned down by the housing supply price elasticity in eq. (21). [Cavalleri et al. \(2019\)](#) estimate this elasticity to be 1.17 for Spain, which corresponds to $\gamma_L = 0.46$.

Without loss of generality we assume $p_h = 1$ in the baseline. This allows us to retrieve two parameters from the model by imposing equilibrium conditions. The first one is the rental intermediary cost, η . Once we set $p_h = 1$, then the after-tax rental price in the baseline economy boils down to rent-to-price ratio. According to the estimates of [Khametshin et al. \(2024\)](#) this ratio is 5.6%. Given that in the Baseline economy, there is a flat housing tax at 0.62% of the house value, we set $p_r = 0.05$ and using equation (23) we solve for η . A similar procedure allows us to directly solve for the amount of land available for production $\bar{L}$. Once the housing price is known, we impose the equilibrium condition in the housing market and solve for $\bar{L}$ needed to exactly offer the demanded quantity in this market from

equation (21). Both of these parameters are then fixed throughout our experiments.

Table 3: Targeted moments. Data vs Baseline values.

	Data	Model
<i>Housing</i>		
Homeownership rate	72.3	72.4
Avg. housing portfolio share	50.0	48.0
Housing holdings Gini	44.1	37.0
<i>Entrepreneurs</i>		
Share entrepreneurs	7.73	7.81
Share of hiring entrepreneurs	55.2	51.9
Entrepreneur’s earnings Gini	49.1	47.3
Share of E among top 1% of income	29.6	30.9
Ratio of median income of E/W	1.33	1.39
Share ent. top 10% wealth	20.1	17.6
<i>Workers and retirees</i>		
Worker’s earnings Gini	40.8	40.6
Homeownership rate 65+	83.8	83.7
Net wealth Gini of retirees	61.3	60.8
<i>Overall economy</i>		
Wealth-to-Income Ratio*	6.7	7.2
Net wealth Gini	68.5	64.2
Top 10% income share	30.5	35.9
Top 1% income share	6.79	6.87

Data source: EFF (2022) with the exception of * which we take from Blanco et al. (2021).

Notice that assuming a unitary price in the baseline only affects the interpretation of our grids: not only the grids are quantities but they reflect the baseline price level. Thus each grid point of $\hat{\mathcal{H}}$ is a quantity and baseline-price adjusted measure of housing units. Nevertheless, this does not affect any of our experiments since the change in p_h will still reflect the change in relative values¹⁰.

The two remaining parameters, h_{min} ($\equiv h_1$) and χ , are calibrated internally, along with ξ , to target the three housing moments in Table 3. In particular, the parameter χ is used to target the share of renters in the economy. Remember that χ represents the utility premium from owning your dwelling. Thus, the higher this value the more incentives there are for the households to become homeowners. The minimum grid size, h_{min} , is used to pin down the distribution of housing assets among the homeowners. Notice that, for example, if h_{min} is set too high the minimum size restriction will be binding for most of the owners, which will lead to a low dispersion.

Notice that a value of $\chi = 1.57$ is large relative to what the literature finds. One reason for this may be the well-documented aspiration of Spanish households to become homeowners¹¹. Beyond this, older households in Spain likely acquired their homes during a period of lower house prices, more generous housing subsidies, and more favorable labor market conditions. Since these conditions no longer prevail—and homeownership rates have been declining in recent years—the estimated preference for homeownership may be upward biased due to cohort effects.

¹⁰On the other hand, assuming a unitary housing price in the baseline saves us a significant amount of computational time. This comes from the two parameters that we solve for (η and $\bar{L}$) and from having one less market clearing condition to satisfy.

¹¹Industry surveys suggest that only around 7% of Spanish households report no interest in owning, roughly half the European average of 15% (Remax, 2025).

4 Benchmark Economy

In this section, we evaluate the benchmark economy’s performance at its initial steady state. Table 3 demonstrates that the targeted moments are broadly well-matched. Overall, our model replicates the central features of the entrepreneurial sector as well as the aggregate homeownership and portfolio shares of the Spanish economy. There are, however, discrepancies concerning wealth inequality, which is under-estimated.

These discrepancies arise due to the introduction of housing. Housing is broadly held across most households and tends to be relatively evenly distributed compared to other assets. Moreover, it represents a sizable share of the total assets in the economy, around 50%. Consequently, the remaining half of households’ wealth in the model has to be extremely unequally distributed (a Gini coefficient around 80) to match aggregate wealth distribution. In reality, the non-housing portion of wealth encompasses diverse assets –including bonds, bank accounts, stocks, and private businesses— each differing in terms of risk and returns. Nevertheless, in our model all households that invest in the financial asset face the same return. Moreover, households use this asset to smooth their consumption.

Table 4: Housing distribution (homeowners only)

	Gini	0-50	50-90	90-99	Top 1%
Data	44.09 ^t	21.36	45.21	25.35	8.08
Model	36.98	20.55	46.83	27.66	4.95

Data source: EFF (2022). *Note:* ^t Targeted.

Having noted these quirks, we also note that our model does a good job in matching the overall wealth and income distributions, as reported in Table 5. The same is true for the housing distribution, reported in Table 4.¹² Additionally, the model performs well in replicating the share of total housing wealth that each income (and wealth) quintile holds. Notice that we do not explicitly target any of these moments from the bivariate distributions, the model delivers a realistic concentration of housing at the top income and wealth brackets. Matching these bi-variate distributions, without targeting them explicitly, reassures us that the model is a good framework to quantify the effects of housing taxation.

¹²We do not report a more detailed distribution due to the discrete nature of housing in our model. We only have 10 grid points for homeowners.

Table 5: Distributions of Income and Wealth

	0–20	20–40	40–60	60–80	80–90	90–99	Top 1%
<i>Wealth</i>							
Data	-0.11	3.69	9.12	17.57	16.18	32.58	20.98
Model	0.84	4.49	8.90	18.21	17.92	37.65	11.99
<i>Housing share by wealth</i>							
Data	1.29	7.49	14.94	23.77	17.63	27.78	7.10
Model	0.53	7.20	13.79	25.73	19.84	27.61	5.29
<i>Income</i>							
Data	5.05	9.90	15.14	22.66	16.78	23.67	6.79
Model	4.83	8.50	13.28	21.69	15.74	29.09	6.87
<i>Housing share by income</i>							
Data	7.75	12.01	16.41	23.87	15.36	20.16	4.44
Model	5.75	12.86	14.80	24.67	17.92	20.97	3.02

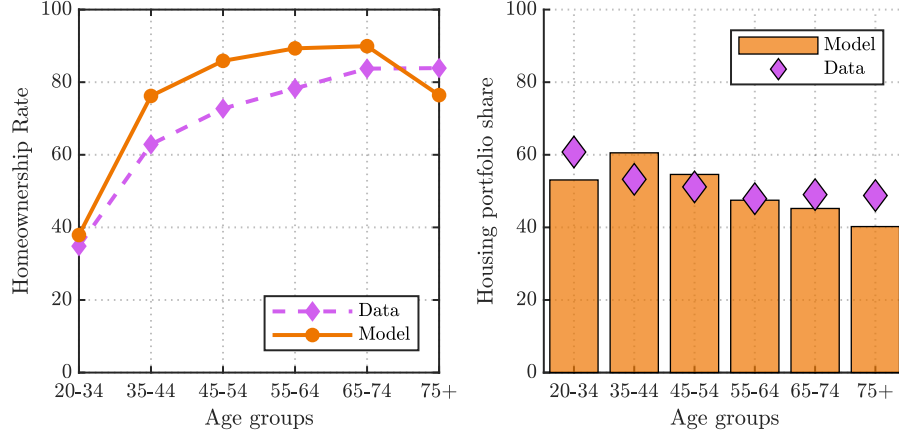
Data source: EFF (2022). *Note:* The share of the top 10% of the income distribution is also targeted. The housing shares by income and wealth represent what share of total housing assets in the economy belong to those particular bins.

As for the other housing moments, we have only targeted the aggregate homeownership rate, the homeownership rate after retirement and the aggregate housing portfolio share. An important question is whether the model replicates the empirical patterns of these moments across different groups. Figure 1 depicts homeownership rates and housing portfolio shares across age, income, and wealth dimensions. Overall, our model captures the essential features of the housing distribution across these three dimensions. That is: 1) homeownership rates increase with income and wealth, and this increase is steeper in the latter case; 2) housing portfolio shares generally decline above the lower-middle income and wealth bins; 3) portfolio shares vary slightly along the life cycle. The fit is not perfect, which is not surprising given that we only targeted the average homeownership and portfolio shares. Regarding homeownership rates, we over-predict homeownership for middle-aged households, and underpredict it for the 75+ years old. Another dimension that is difficult to capture is the portfolio shares at the very top of both income and wealth. Notice that in the data the housing portfolio shares at the top 1% decrease steeply. The reason is that those households invest in other assets that we are not considering here, such as shares or private businesses¹³.

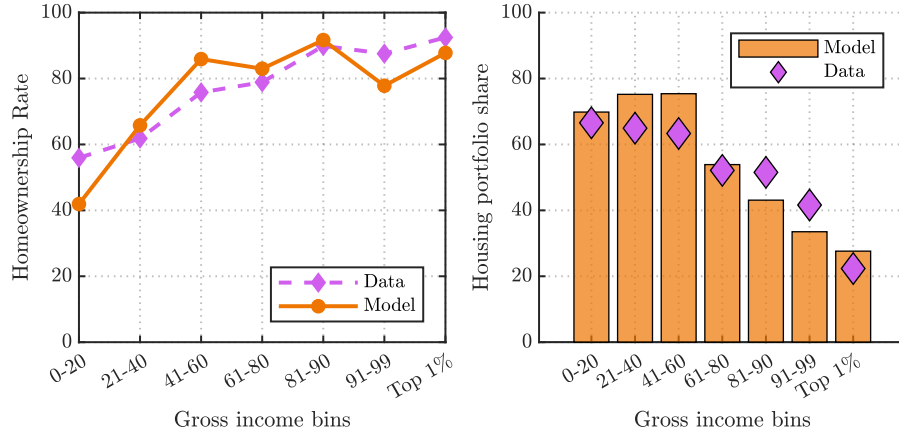
Another set of fully untargted moments concerns housing portfolio shares by occupation. Table 6 reports the housing portfolio shares of workers and entrepreneurs across net-wealth bins. The model captures the broad pattern that workers allocate a larger share of their wealth to housing than entrepreneurs do. For the 20th–80th percentiles of the wealth distribution, it replicates the housing shares of both workers and entrepreneurs reasonably well, though it underpredicts the housing share of workers at the bottom of the distribution. At the top of the distribution, however, the model overpredicts the housing share of entrepreneurs and underpredicts that of workers, so that the two occupations hold similar housing shares (e.g. 24.5% for entrepreneurs and 23% for workers, versus 10% and 35%, respectively, in the data for the top 1%). This shortfall reflects the model’s abstraction from features that would deliver more realistic portfolios at the top, such as equity holdings or a distinction between the market value of a business and the market

¹³If we compute housing shares without taking the value of private businesses into account, then our estimates are even more closely aligned since the top portfolio shares become 25% and 23% for income and wealth respectively.

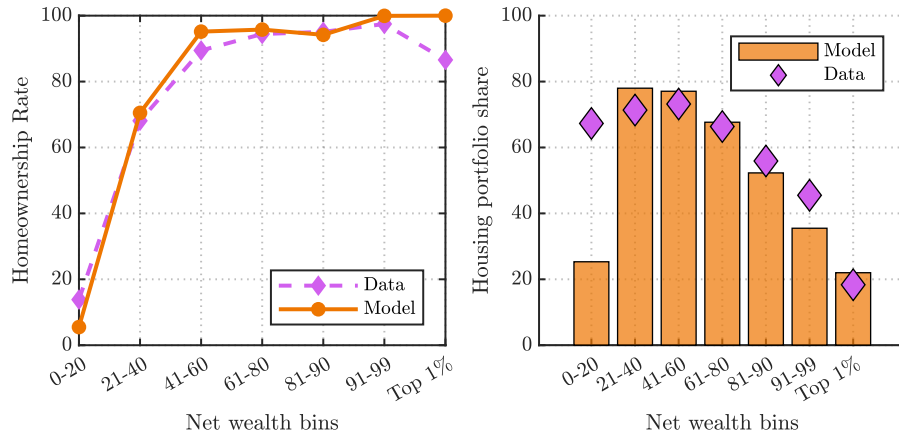
Figure 1: Homeownership rates and portfolio shares across different dimensions.



(a) By age groups



(b) By income percentile bins



(c) By wealth percentile bins

value of its capital net of debt. If anything, a closer match to the data would strengthen our results: as we argue in section 5.1, shifting the tax system toward progressive housing taxation and away from progressive income taxation is

more desirable the larger the gap between workers' and entrepreneurs' housing portfolio shares.

Table 6: Housing portfolio share by occupation and wealth bin

	0–20	20–40	40–60	60–80	80–90	90–99	Top 1%
<i>Workers</i>							
Data	73.1	73.1	75.1	68.8	54.4	51.2	35.4
Model	13.4	77.3	74.7	66.6	57.0	37.4	23.0
<i>Entrepreneurs</i>							
Data	–	41.8	56.6	45.2	44.7	33.5	10.3
Model	–	56.8	63.6	45.1	27.5	40.6	24.5

Data source: EFF (2022). *Note:* for entrepreneurs, the 0–20 wealth bin is omitted because it contains too few observations in the EFF; the corresponding model moment is also omitted for comparability.

Nevertheless, the model's ability to reproduce several untargeted features of the data, such as the joint distribution of housing, income, and wealth, as well as life-cycle and cross-sectional patterns in ownership and portfolio shares, further strengthens our confidence in its structure. This good fit along non-targeted dimensions suggests that the model captures the key mechanisms shaping housing decisions and can therefore be reliably used to study housing taxation.

5 Main Experiments

After showing that the model economy replicates the empirical distributions of wealth, housing and income, we analyze whether a change in the housing taxation system might be welfare-enhancing. In order to compute optimal policy rules, we have to take a stand on a welfare measure. As a measure of optimality, we use the expected value function of newborn agents. This is a reasonable optimality measure because our experiments are steady-state comparisons. In effect, we ask whether a newborn agent would prefer to live in the baseline economy or in an alternative one with a different tax structure. To make quantitative comparisons on welfare, we define a consumption-equivalent measure as:

$$\mathbb{E} [\mathcal{U}^*(\{\mathbf{c}^*, \mathbf{d}^*\})] = \mathbb{E} [\mathcal{U}(\{(1 + \bar{c}e)\mathbf{c}, (1 + \bar{c}e)\mathbf{d}\})] \quad (29)$$

where $\mathbb{E} [\mathcal{U}^*(\{\mathbf{c}^*, \mathbf{d}^*\})]$ denotes the expected lifetime utility of a newborn under the counterfactual policy $(\tau_h^{l*}, \tau_h^{p*}, \tau_y^{l*}, \tau_y^{p*})$, with $\{\mathbf{c}^*, \mathbf{d}^*\}$ the associated allocations of non-durable consumption and housing services. $\mathbb{E} [\mathcal{U}(\{\mathbf{c}, \mathbf{d}\})]$ denotes the corresponding quantity in the baseline economy. The welfare measure $\bar{c}e$ is thus the proportional increase in both non-durable consumption and housing services that would make a newborn indifferent between the baseline and the counterfactual economy.

All of the considered fiscal reforms are budget-balanced. Note that a balanced-budget reform is not necessarily revenue-neutral, due to the presence of pension payments that are proportional to the endogenous wage rate. Thus, any reform that increases (decreases) wages will require higher (lower) revenues than in the baseline.

We next examine several alternative counterfactuals. Section 5.1 analyzes the optimal housing tax schedule in our economy, highlighting the role of progressivity and contrasting housing and income tax schedules. Finally, Section 5.2 presents the joint optimization of housing and income taxation to simulate the overall optimal fiscal system.

5.1 Optimal housing vs income taxation

We begin by comparing the optimal housing and income tax schedules. Progressive income taxation is the primary redistributive tool in most economies and has been widely studied in the literature. Therefore, it constitutes a natural benchmark for assessing the redistributive and efficiency implications of progressive housing taxation. To ensure a fair comparison, we do not contrast the existing income tax schedule with the optimal housing tax design. Instead, we compare the effects of both tax schedules at their respective optimal structures.

We define optimal housing taxation as the problem of maximizing the expected utility of a newborn at steady state with respect to the housing tax level (τ_h^l) and progressivity (τ_h^p) parameters, subject to the government budget constraint. Note that this tax is effectively a tax on housing consumption, as it is paid by both renters and homeowners (see eq. (8)). Analogously, optimal income taxation maximizes the expected utility of a newborn with respect to the income tax level (τ_y^l) and progressivity (τ_y^p) parameters, subject to the same constraint. We refer to these two experiments as *Opt H* and *Opt Y*, respectively.

Throughout this section, we permit negative tax rates, which arise naturally within the parametric tax function adopted in our framework. In section E.1 of the Appendix, we present the corresponding results under the additional constraint that subsidies are ruled out.¹⁴ Under this restriction, progressive housing taxation proves considerably more potent than the progressive income one, as it improves the welfare of the poorest households through equilibrium effects, something that progressive income taxation is unable to achieve. We now compare the two optimal schedules along several dimensions; Table 7 presents the detailed results.

Tax Rates: As shown in the third column, the optimal housing tax schedule features progressivity and a higher average effective tax rate. The average tax rate is 2.48%, and 59% of households are effectively paying this tax, with the 0-tax threshold being at roughly the 30th percentile of the baseline’s housing size distribution. The progressivity of optimal housing taxation stands at 0.06, less than two-thirds of the progressivity parameter of the current income tax.

The optimal income taxation is highly progressive (at 0.571, about 6 times more than current progressivity), and the average tax rate is also about 50% (6.6 p.p.) larger than the baseline economy (see fig. 2). About 35% of households pay this tax, while the rest receive subsidies. Both Opt Y and Opt H feature considerably larger redistribution than the baseline, especially the Opt Y one, for which the Gini of consumption drops by a third. Given that both tax schedules are highly progressive, it is worth remembering that the model does feature financially constrained entrepreneurs, but no endogenous labor supply or human capital accumulation, which would impose a further limit on the optimal progressivity of both the income and housing tax schedules.

Resource Allocation and Prices: The two tax schedules have distinct effects on prices and the allocation of the economy’s resources between capital and housing. The optimal progressive income taxation causes a large drop in the stock of capital (by 27%), while leaving the housing stock 10% lower. The lion’s share of the fall in the capital stock comes from the self-employed, whose demand collapses by 2/3, which aligns with the expected result of a highly progressive income tax on entrepreneurs. In this economy the production of the consumption good drops by 11%, interest rates rise to 6.2%, and wages fall by 9%. These results are consistent with the broader literature covering the effect of higher income tax progressivity, see Brüggemann (2021) or İmrohoroglu et al. (2023). In our setting, the optimal income policy also leads to a lower housing demand due to income shrinking, resulting in housing price decreasing by 19%.

¹⁴Specifically, when subsidies are not permitted, the tax functions take the form $t_g(g) = \max\{0, (1 - \tau_g^l(g/\bar{g})^{-\tau_g^p})g\}$ for $g \in \{y, h\}$; otherwise, they take the unconstrained form $t_g(g) = (1 - \tau_g^l(g/\bar{g})^{-\tau_g^p})g$.

Table 7: Housing vs Income Taxation Results

	Baseline	Income	Housing	Joint
<i>Tax Parameters</i>				
τ_y^l Income Tax Level	0.882	0.679	0.882	0.41
τ_y^p Income Tax Prog	0.1	0.571	0.1	0
τ_h^l H Tax Level	0.994	0.994	0.972	1.103
τ_h^p H Tax Prog	0	0	0.059	0.151
<i>Tax Structure</i>				
Effective income tax rate (%)	12.8	19.37	12.77	59
Share of hhs paying income tax	72.16	35.33	71.86	100
Effective housing tax rate (%)	0.62	0.62	2.48	-11.38
Share of hhs paying housing tax	100	100	59.23	0.09
Gini of Total Consumption*	0.39	0.26	0.32	0.23
<i>Prices</i>				
p_h Housing price	1	0.81	0.77	0.73
p_r Rental price	0.05	0.06	0.03	0.04
r Interest rate (%)	4.09	6.23	3.22	5.14
w Wage	1	0.91	1.05	0.95
<i>Aggregate Variables</i>				
Production	1	0.89	1.06	0.96
Financial wealth	1	0.61	0.95	0.71
Capital	1	0.73	1.15	0.85
Housing Stock	1	0.9	0.69	0.75
Rental Stock	1	0.21	0.35	0.29
Share of homeowners	72.4	92.07	86.36	90.76
Welfare ($\Delta\%$)	-	11.9	9.96	20.43

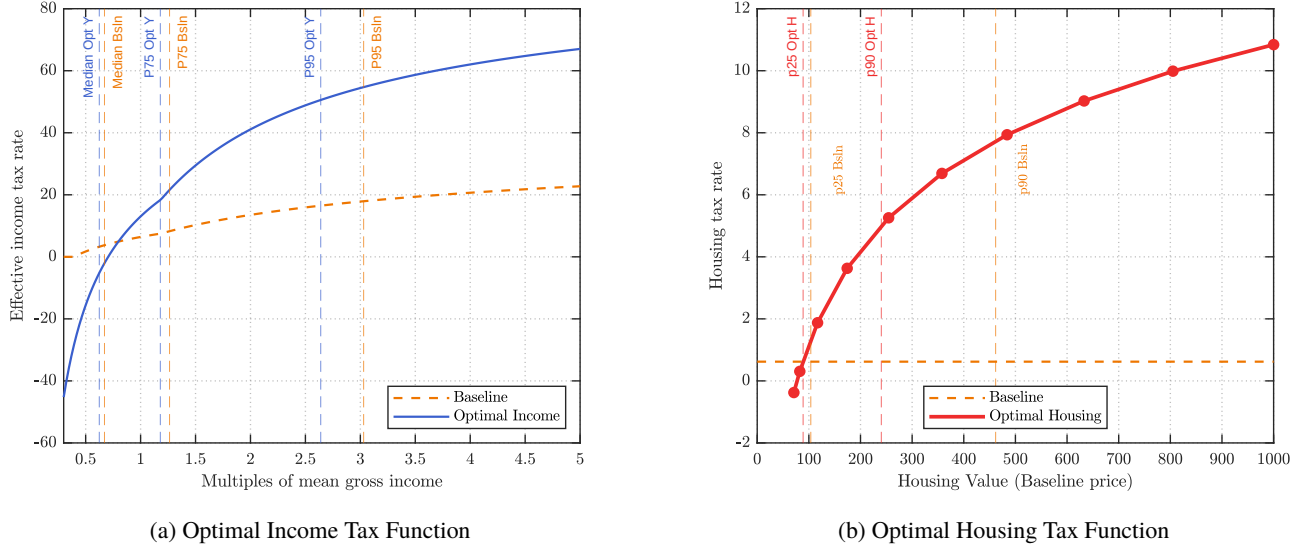
* Total consumption is defined as $C_{tot} = c^\xi d^{1-\xi}$

In contrast to *Opt Y*, the optimal housing tax induces a reallocation of the economy's resources from housing to capital. Specifically, *Opt H* reduces the demand for housing and generates a large drop in the housing stock by 31% with respect to the baseline, as well as a drop in the housing price by 23%. Since the increase in tax rates for large houses reduces the opportunity cost of investing in the financial asset for rich households, the interest rate goes down, and the capital stock in the economy rises by 15% (fig. 3). The capital deepening leads to an increase in wages by 5%. The production of the consumption good also increases in this case by 6%.

Crucially, the 15% increase in the capital stock under *Opt H* masks differentiated effects on the capital employed by the corporate sector (increases) and the self-employed (stable). To understand why this happens, in Table 8 we provide additional statistics on self-employed households across counterfactuals.

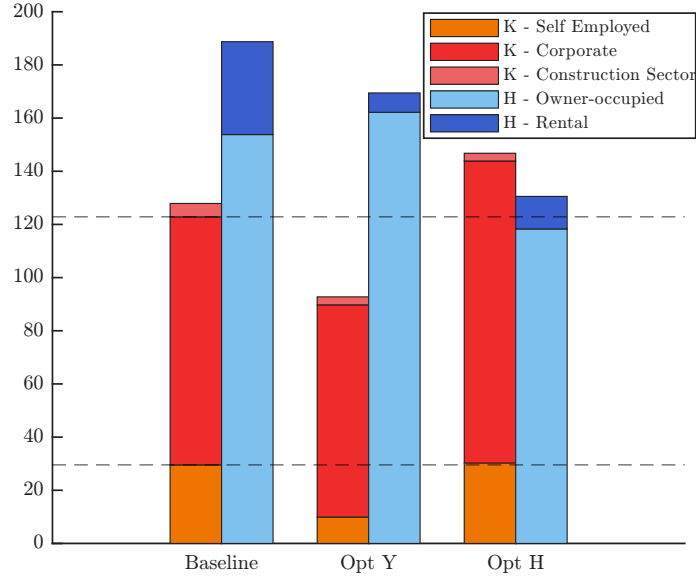
The first thing to note is that under *Opt H*, lower house prices mechanically reduce the collateral value of a given housing position. However, this force is partly offset by the portfolio response of entrepreneurs. Progressive housing taxation makes large housing positions more expensive and induces wealthy households to shift their portfolios toward the financial asset, which is more effective as collateral. At the same time, the reform changes the composition of the self-employed: the share of self-employed households falls from 7.8% to 7.1%, while their average entrepreneurial ability rises by about 3%.

Figure 2: Optimal Tax Functions: Optimal Income vs. Optimal Housing tax schedules



Note: this figure plots the optimal tax functions described in section 5. Panel (a) plots the optimal income tax function (*Opt Y* in the main text) and panel (b) describes the optimal housing tax schedule (*Opt H*). The circles denote tax rates at the housing grid points.

Figure 3: Capital and Housing Demand



Note: Housing is evaluated in constant (baseline) prices ($p_h = 1$) across counterfactuals.

As a result, the remaining self-employed are both more productive and, because the interest rate is lower under *Opt H*, would like to operate at a larger scale. Their average unconstrained capital demand rises by 17% relative to the baseline. Average collateral also increases, but by a smaller amount, 8.9%, reflecting the portfolio shift toward financial assets. Since desired scale increases more than collateral, the share of constrained self-employed households rises from 58.6% to 63.7%.

When comparing the effects of the *Opt Y* and *Opt H* reforms on self-employed the contrast is stark. Increasing redis-

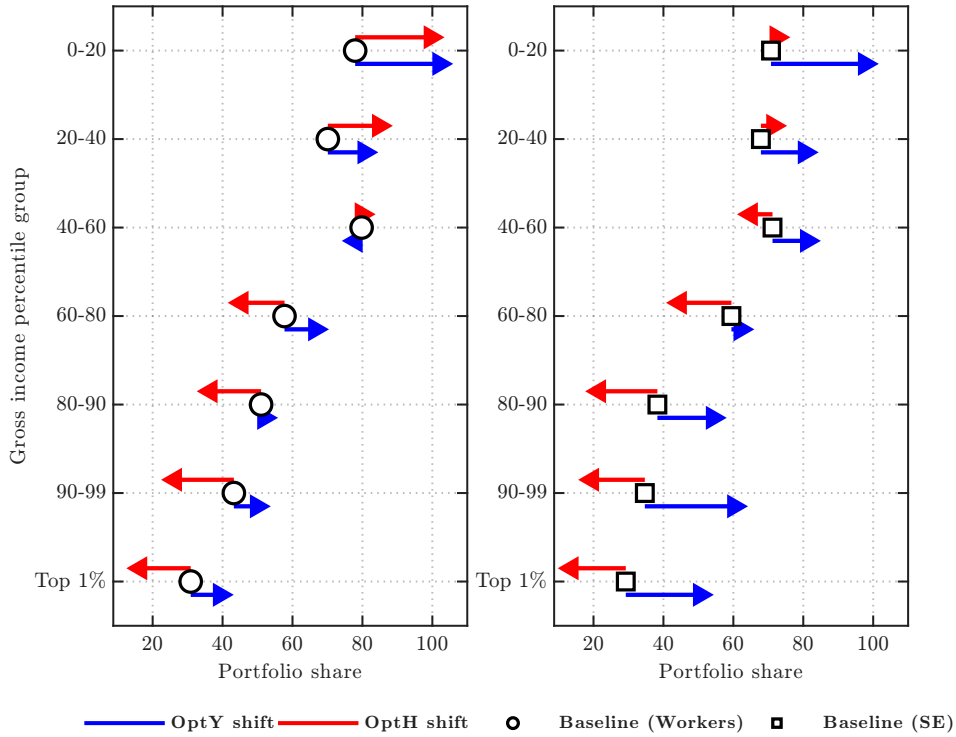
Table 8: Self-Employed: The Role of Collateral

	Baseline	Opt Y	Opt H
Mean k^*	100	33.3	113.2
Mean unconstrained k^*	100	79.1	117
Average collateral	100	33.0	108.9
Share of S.E.	7.8	7.9	7.1
Share constrained S.E.	58.6	65.5	63.7
Mean entrepreneurial ability	100	99.6	103.1
S.E. wealth share	12.5	7.4	14

Note: Mean k^* , mean unconstrained k^* , average collateral, and mean entrepreneurial ability are normalized to 100 in the baseline. All other rows report percentages.

tribution through income taxation drives down both the desired level of capital for the most productive self-employed and their value of the collateral by a wide margin. A crucial feature of the progressive housing tax reform is that it taxes the most productive entrepreneurs less than the least productive, after conditioning on wealth, allowing them to accumulate larger collateral and operate at a larger scale.

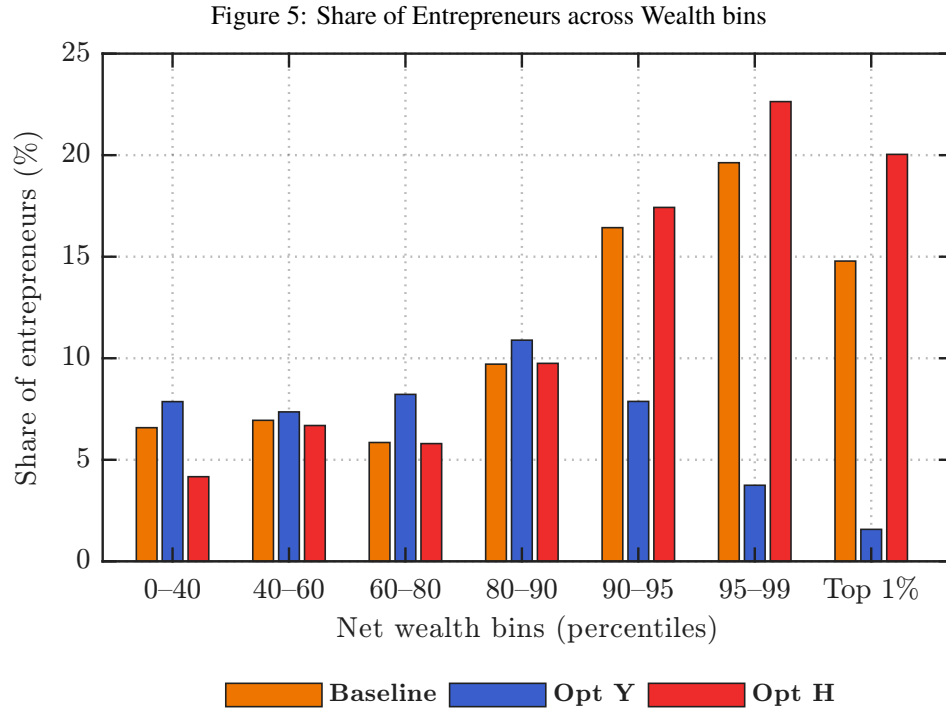
Figure 4: Portfolio shares by income bins



We can see this differential treatment by looking at the effect of the considered tax reforms on the portfolio shares. Figure 4 shows that progressive housing taxation generates distinct incentives across the income distribution. Under the *Opt H* scenario, low-income households increase both their absolute and relative holdings of housing, while the steep tax rates reduce the housing share among the richest groups. Notably, the *Opt Y* reform leads the richest entrepreneurs to increase their housing portfolio shares substantially more than workers in the same income bin; the opposite holds under the *Opt H* reform, which induces the self-employed to reduce their housing portfolio shares by more, or increase them by less, relative to their worker counterparts. Consequently, the *Opt H* policy yields a more efficient allocation of

wealth accumulation, as households with higher entrepreneurial ability reallocate their wealth toward financial assets (see also Appendix table A1).

At this point it is instructive to interpret our results through the lens of the “use-it-or-lose-it” mechanism described in (Guvenen et al., 2023). In our model, for a given level of wealth, households face a portfolio choice between two assets: housing, which yields a homogeneous utility return, and financial assets, on which households earn heterogeneous returns depending on their wealth level and entrepreneurial ability.¹⁵ Because returns on financial assets vary with entrepreneurial ability, heterogeneous returns on wealth arise across households. Guvenen et al. (2023) show that in such an environment there is an incentive to shift taxation from capital income to wealth: while capital income taxes place the burden on the highest-ability entrepreneurs, wealth taxation reallocates wealth toward the most productive ones. In our environment, for a given level of wealth, housing taxation imposes a higher burden on households with a larger housing share in their portfolio—that is, households with lower entrepreneurial ability and lower returns on wealth. This mechanism is an amplified analogue of the use-it-or-lose-it channel of Guvenen et al. (2023): households with higher returns on wealth pay less in housing taxes than households of equal wealth but lower entrepreneurial ability¹⁶. Consistent with this mechanism, the share of the self-employed in the top 1% of the wealth distribution rises from 15% in the baseline to 20% under *Opt H*, while collapsing to 1.6% under *Opt Y*; similar patterns hold across other top wealth bins (see fig. 5).

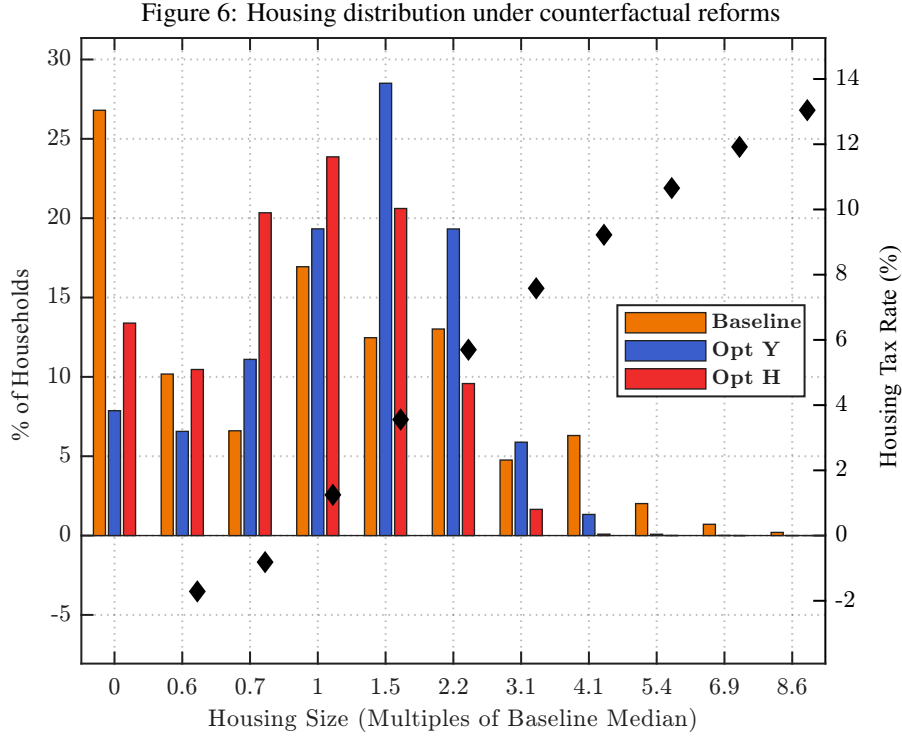


Housing Distribution: Although both tax reforms squeeze the right tail of the housing distribution, the mechanics differ. Under *Opt Y*, the high effective income taxes at the top decrease their ability to purchase larger housing units (fig. 6). Meanwhile, *Opt H* makes it prohibitively expensive to live in those large dwellings, without a direct tax that lowers the disposable income of the households at the very top.

¹⁵There is some heterogeneity in the utility returns from housing arising from the equivalence scale and warm-glow bequests, both of which depend on age; however, this heterogeneity is limited relative to that stemming from entrepreneurial ability.

¹⁶As discussed in section 4, the model fails to generate a meaningful wedge between the housing portfolio shares of workers and entrepreneurs at the top of the wealth distribution, as observed in the data. A better fit on this dimension would only strengthen the mechanism and results discussed here; For housing shares along the wealth distribution and across occupations see also fig. A2 in the Appendix.

Changes also occur at the lower end of the distribution. In particular, both reforms lead to a sizeable increase in homeownership rates. Under *Opt H*, the homeownership rate rises by about 14 percentage points, while under *Opt Y* the increase is close to 20 percentage points. Two mechanisms contribute to this result. First, the reforms introduce transfers through negative tax rates for low-income households, which relax liquidity constraints and allow younger or poorer households to accumulate the down payment required for homeownership. Second, both policies reduce equilibrium house prices, lowering the entry cost of purchasing a home. As a result, households that would otherwise remain renters can transition into homeownership earlier in the life cycle, smoothing their housing consumption over time.



Note: The graph presents the housing distribution (left axis) and the corresponding tax rates under *Opt H* (right axis) for each housing size. The x-axis reports the housing sizes as multiples of the Baseline median one. The first bin (0 housing value) corresponds to renting. Reported tax rates are percentage shares of housing values. For Baseline and *Opt Y* the housing tax is flat and equal to 0.62%.

Effects on Consumption: Both tax schedules yield significant welfare gains with respect to baseline, 11.9% and 10% under the optimal income and housing tax, respectively. To understand the source of the welfare gains we perform a variance decomposition of total consumption C_{tot} , which we define as $c^\xi d^{(1-\xi)}$, to uncover consumption smoothing effects between and within ages. In table 9 we present the results. Both policies reduced the mean of C_{tot} by 9% and 8%, respectively. It is worth pointing out that under the *Opt H* scenario the average non-durable consumption actually increases by 6%. Nevertheless, the fall in the consumption of housing services is much stronger under this scenario (-28%) than compared with the *Opt Y* counterfactual (-5%).

As for the variance, both policies lower it substantially: by 57% in the case of the *Opt Y*, and 35% in the case of the *Opt H* tax. *Opt Y* is very successful in reducing the within-age variance (by more than two-thirds, third column of table 9), but it fails to affect the between-age variance. On the other hand, *Opt H* reduces by a third the within-age variance of C_{tot} and by half the between-age part.

A key feature of the model that generates this discrepancy is that income and consumption are not perfectly correlated in the model. Agents earn higher incomes when young, but choose to smooth consumption over the life-cycle. As fig. 7

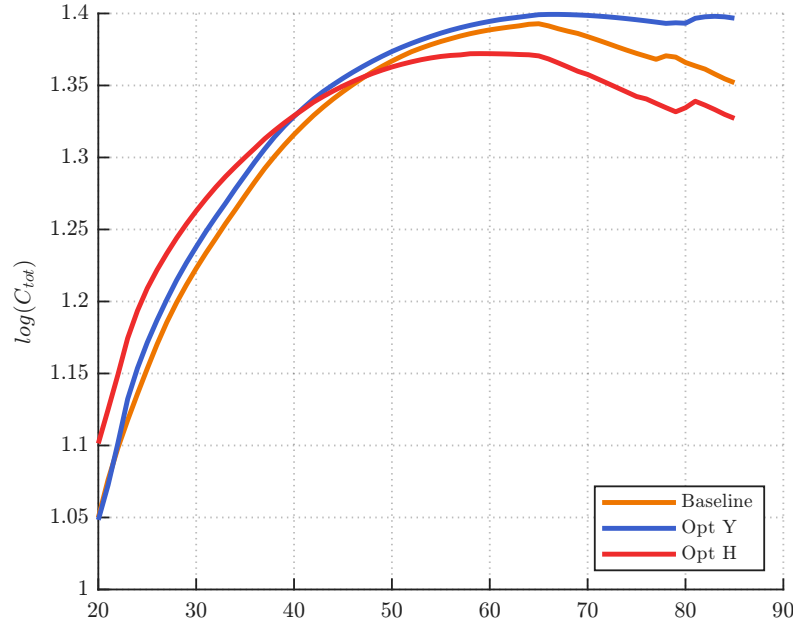
Table 9: C_{tot} variance decomposition

	Mean	Total variance	Within-age	Between-age
Baseline	100	100	81	19
<i>Opt Y</i>	91	43	23	20
<i>Opt H</i>	92	65	55	10

Note: Total consumption is defined as $C_{tot} = c^\xi d^{1-\xi}$. The first column reports $E[C_{tot}]$, normalized by its baseline value. The last three columns decompose $Var(\log C_{tot})$ using $Var(\log C_{tot}) = E[Var(\log C_{tot} | j)] + Var(E[\log C_{tot} | j])$ where j denotes age. Total variance and its within- and between-age components are normalized by baseline total variance.

shows, baseline C_{tot} grows at a fast pace until the age 50, and is close to flat afterwards. Housing consumption has the same pattern. The *Opt Y* tax fails to smooth C_{tot} over age much, as it taxes high-earning working-age households, and redistributes to low-income, but potentially wealthy in terms of consumption, retirees. On the other hand, housing taxation, as every consumption tax would do, successfully smoothes C_{tot} over the life-cycle, by taxing consumption-rich retirees and subsidizing consumption-poor young. Notice also that C_{tot} doesn't account for equivalence scales, and household size decreases at retirement, amplifying the importance of this channel for welfare.

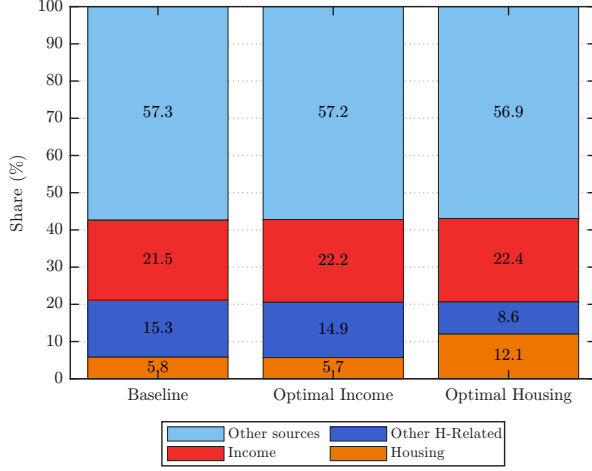
Figure 7: Mean total consumption across counterfactuals



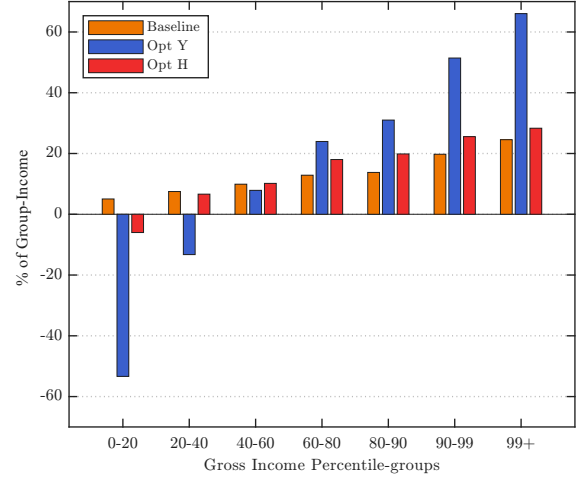
Note: The graph plots mean total consumption (C_{tot}), defined as $c^\xi d^{(1-\xi)}$, with the vertical axis in log scale.

Government Revenues and Tax Burden: The two taxation plans modify the structure of government revenues in different ways. The *Opt Y* tax barely affects the structure of Government revenues, implying that most of the extra income revenues from higher progressivity, are given as transfers. On the other hand, under *Opt H* tax schedule, the housing tax revenues expand to 12.1% of total government revenues, but this does not happen at the expense of the income tax revenues, which slightly rise. Instead, the shares that drop are housing-related taxes, such as inheritance, transaction taxes, and the profits of the construction firm, all affected by the lower housing stock and the lower price.

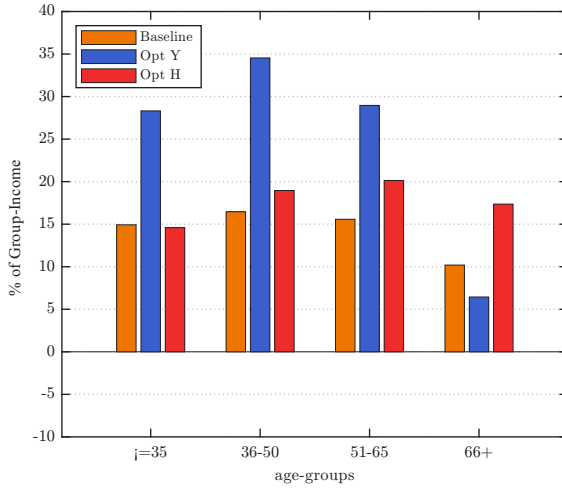
Figure 8: Comparison of revenue shares and tax burdens under *Opt Y* and *Opt H* reforms



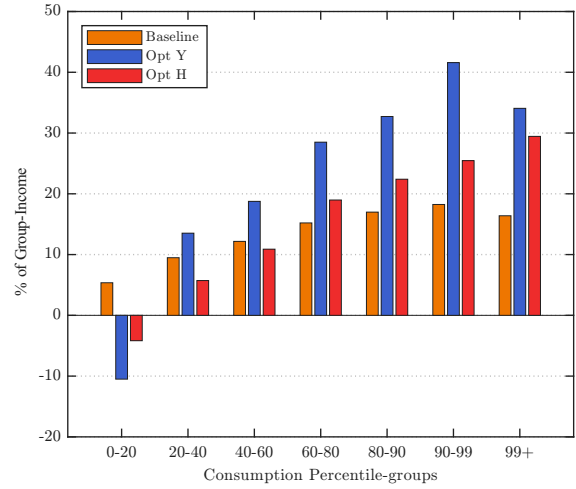
(a) Comparison of governments' revenue shares



(b) Tax burden by income bins



(c) Tax burden by age groups



(d) Tax burden by C_{tot} bins

Note: Panel (a) reports the composition of government revenue by source, as a share of total revenues net of subsidies. “Other H-Related” comprises the property transaction tax, the inheritance tax, and the profits of the construction firm. “Other sources” comprises social security contributions and the consumption tax. Income and housing tax revenues are reported net of the associated subsidies. Panels (b)-(d) split the population into income, age, or consumption bins. For each bin, they divide aggregate income and housing taxes by aggregate gross household income. For consumption-bins total consumption $C_{tot} = c^\xi d^{(1-\xi)}$ is used

They also differ on who bears the tax burden, as the *Opt Y* reform's taxes are primarily paid by income-rich working-age households, while *Opt H* also taxes significantly retirees. To show this, we categorize households by income bins, (total) consumption bins and age groups, and for each bin we aggregate income and housing taxes paid, and calculate the total housing and income tax burden that this group pays as a fraction of their income. We plot the results in fig. 8 (b)-(d). As expected, *Opt Y* taxes heavily the top of the income distribution (with the joint burden of housing and income taxation being 66% of income for the top 1%), and provides large subsidies to the bottom 20% (53%). In contrast, *Opt H*'s tax burden for the top 1% is 28%, which is only slightly higher than baseline's 24%, and much lower than *Opt Y*'s. Subsidies to the poorest 20% are also much lower than in *Opt Y*, at 6%.

In terms of total-consumption bins, both tax schedules impose a substantial burden on the top of the distribution: the top 1% faces a tax rate of 34% under *Opt Y* and 29% under *Opt H*. The large discrepancy between the tax burden paid by the top 1% of the income distribution and the top 1% of the consumption distribution indicates that these groups consist of different households; the highest earners are not necessarily the highest spenders. In fact the *Opt Y* tax schedule turns out to be regressive at the very top of the total-consumption distribution. A similar pattern appears at the bottom of the distribution. Under *Opt Y*, the subsidies granted to the consumption-poorest 20% amount to only 10% of the bin's income, far below the 53% calculated for the bottom 20% of the income distribution. It is also notable that *Opt H* delivers substantially lower subsidy rates to the bottom 20% of both the income and consumption distributions compared with *Opt Y*. Thus, under the *Opt H* reform, the welfare gains of the poorest households arise largely from equilibrium effects—lower house prices and higher wages—rather than from housing subsidies alone.

Where the two reforms differ the most is on the tax burden they impose across age groups (fig. 8c). *Opt Y* reform significantly increases the tax burden to working age households (from 16% to 35% for the 36-50 bins and from 16% to 29% for the 51-65 bin), while it shrinks it for retirees; as many retirees are income-poor they earn subsidies and the 66+ bin contributes in net only 6% of bin's income. On the other hand, the *Opt H* tax plan, increases significantly the tax burden paid by retirees, from 10% to 17%, while keeping it stable for those below 35 (at 15%). This way the *Opt H* reform increases the tax burden not on young households but mainly on income-poor but consumption-rich retirees, which is crucial since the average household consumes more in retirement than under the age of 35, as shown in fig. 7.

Welfare Gains/Losses across ability: In order to uncover the winners and losers from each reform, we calculate the welfare gains of newborns of different labor and entrepreneurial ability (table 10). Both reforms reduce the expected welfare of high labor and entrepreneurial ability newborns, and increase that of low labor and entrepreneurial ability newborns. However, the distribution of welfare gains is much more polarized under *Opt Y* reform, ranging from gains of 61% for the households with the lowest realizations of z and θ , to losses of 55% for the households with the highest realizations of both. Under *Opt H*, welfare gains range from -20% to 34%. Another crucial difference, is how the "middle-class" of labor ability (z_4, z_5) fares across reforms, especially for newborns of the lowest entrepreneurial ability. Under *Opt Y*, they suffer losses of -6% and -22% due to the high progressivity of income taxation, while under *Opt H* the newborn with skills (z_4, θ_1) gains 0.5% in consumption equivalent and the one with skills (z_5, θ_1) faces losses of only -8%. Newborns born in these states constitute a sizable 25.6% of each cohort's population. Overall, a majority of 66% of each newborn's cohort would prefer living in the *Opt Y* economy instead of in the baseline, while the *Opt H* economy is preferred by 89% of newborns.

(a) Optimal income tax (<i>Opt Y</i>)					(b) Optimal housing tax (<i>Opt H</i>)				
	θ_1	θ_2	θ_3	θ_4		θ_1	θ_2	θ_3	θ_4
z_1	60.6 (5.0)	60.4 (0.6)	56.1 (0.1)	-1.2 (0.3)	z_1	33.8 (5.0)	33.1 (0.6)	31.4 (0.1)	18.1 (0.3)
z_2	32.4 (20.6)	31.8 (2.4)	25.5 (0.2)	-29.6 (1.1)	z_2	18.5 (20.6)	18.0 (2.4)	15.8 (0.2)	-2.9 (1.1)
z_3	11.5 (32.4)	10.8 (3.8)	4.1 (0.4)	-42.4 (1.7)	z_3	8.7 (32.4)	8.2 (3.8)	5.5 (0.4)	-11.7 (1.7)
z_4	-5.5 (20.6)	-6.1 (2.4)	-11.5 (0.2)	-49.4 (1.1)	z_4	0.5 (20.6)	0.2 (2.4)	-2.1 (0.2)	-16.3 (1.1)
z_5	-21.7 (5.0)	-22.1 (0.6)	-25.9 (0.1)	-54.0 (0.3)	z_5	-8.1 (5.0)	-8.3 (0.6)	-9.6 (0.1)	-19.1 (0.3)
z_6	-22.7 (0.9)	-23.1 (0.1)	-26.4 (0.0)	-55.0 (0.0)	z_6	-10.3 (0.9)	-10.4 (0.1)	-11.2 (0.0)	-19.6 (0.0)

Note: Entries report consumption-equivalent welfare gains for newborn households, in percent. The cohort share of households for each combination of labor productivity z and entrepreneurial ability θ is reported in parentheses.

To sum up, progressive housing taxation facilitates welfare redistribution through channels distinct from those of income tax progressivity. By raising the opportunity cost of housing investment for wealthy households, it triggers two key effects. First, the reduction of the demand for housing among the rich leads to lower house prices, thereby improving affordability for poorer and younger households—especially when complemented by housing subsidies. Second, as affluent households reallocate their portfolios from real estate to financial assets, aggregate capital rises. This capital deepening results in higher wages and lower interest rates. Crucially, this mechanism is especially potent in the presence of entrepreneurship, as entrepreneurs exhibit higher returns on wealth and higher saving rates.

Unlike income taxation—which applies uniform rates to income-rich workers and entrepreneurs—progressive housing taxation implicitly favors entrepreneurs by taxing them less at equivalent income and wealth levels, due to their lower housing exposure. High-ability entrepreneurs, who also save at higher rates, can expand their business, raising aggregate production and wages. In doing so, it mitigates the trade-off inherent in progressive income taxation, which tends to suppress capital accumulation and thereby dampen output and wages.

Moreover, housing taxation proves more effective in smoothing consumption across the life cycle. Since income and consumption are imperfectly correlated—many retirees are consumption-rich but income-poor—income taxation is a weak tool for redistributing welfare from old to younger cohorts. In contrast, progressive housing taxation achieves this goal by making it costly to retain large homes for income-poor retirees, thereby reallocating housing resources toward younger, liquidity-constrained households.

5.2 Optimal Joint Housing and Income Taxation

Having presented the distinct macroeconomic effects of optimizing the housing and income tax schedules separately, we now turn to the jointly optimal design of income and housing taxation. Specifically, we optimize the tax system over four parameters: the level and progressivity of both income and housing taxes $(\tau_y^l, \tau_y^p, \tau_h^l, \tau_h^p)$, with $\tau_y^p, \tau_h^p \in [0, 1]$. As in the previous section, we allow for subsidies in both income and housing tax schedules and restrict our attention to budget-neutral reforms. The objective remains to maximize the expected lifetime welfare of a newborn in steady state. We denote this counterfactual by *Opt Joint*.

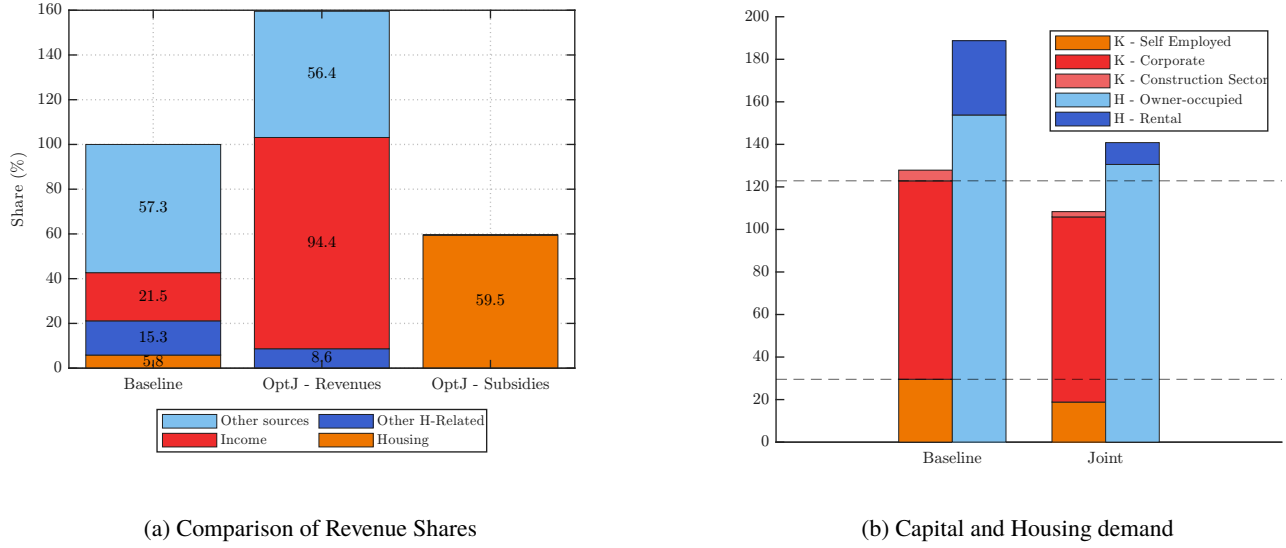
Table 7, 4th column, presents the optimal tax parameters together with equilibrium prices and aggregates. What stands out is the nearly flat and sizable income tax, under which all households contribute 59% of their earnings. Interestingly, this elevated uniform rate remains below the tax rates faced by the top 1% of earners under the *Opt Y* reform (see fig. 8b). Redistribution is implemented entirely through a steeply progressive housing tax schedule, whereby 99% of households receive subsidies, with an average effective subsidy rate of 11%.¹⁷

The net housing subsidies account for a massive 60% of (not subsidies-related) government expenditures, and they are financed mainly by the flat income tax which stands at 94% of government expenditures (see fig. 9a). The steepness of the tax schedule prevents households from living in houses much larger than the 0-tax threshold, which corresponds to the 86th percentile of the housing size distribution in the Baseline economy (fig. 10). The policy exerts effects on prices that are a mix of the two separate policy changes. The housing price falls more than under *Opt Y*, and *Opt H*, while interest rate rises moderately to 5.1%, and the wage rate falls by 5%.

Figure 9b plots the stocks of housing and capital in the economy. The housing stock decreases by 25%, and the capital stock decreases by 15%, about half of the drop under *Opt Y*. The decline in self-employed capital, at 35%, is substantial but considerably more moderate than the 66% collapse under *Opt Y*.

¹⁷The precise income and housing tax functions are depicted in fig. A1, in the appendix.

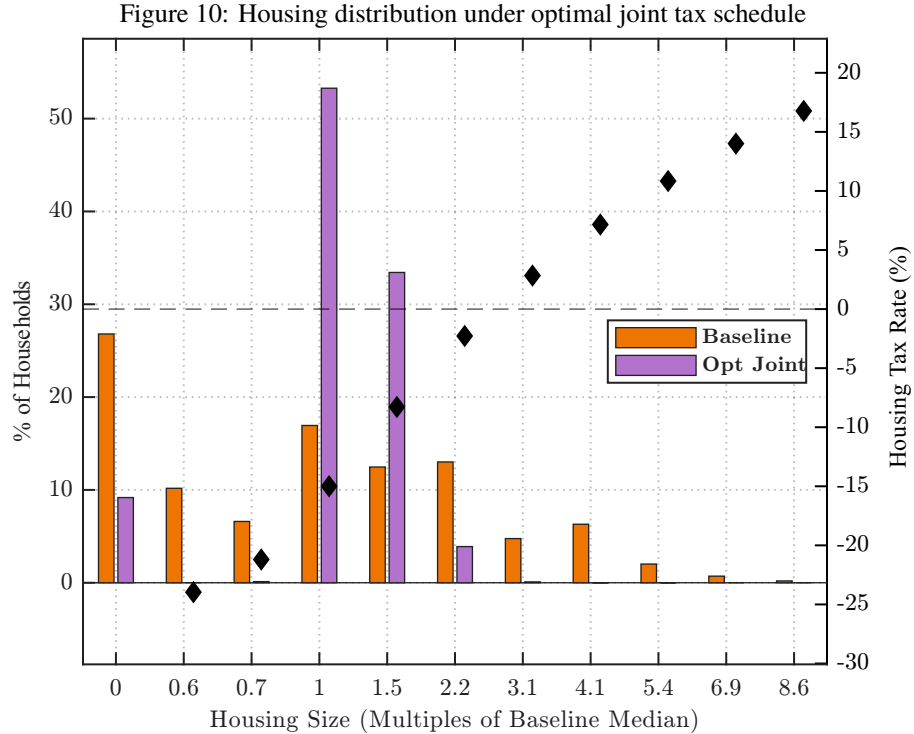
Figure 9: Optimal Joint Housing and Income Taxation:
Government Revenues and Resources Allocation



(a) Comparison of Revenue Shares

(b) Capital and Housing demand

Note: Panel (a) reports the composition of government revenue by source, as a share of total revenues net of subsidies. “Other H-Related” comprises the property transaction tax, the inheritance tax, and the profits of the construction firm. “Other sources” comprises social security contributions and the consumption tax. Only the income and housing taxes generate subsidies; each is reported net of them, appearing in the revenues column if net revenue is positive and in the subsidies column otherwise. Panel (b) reports the housing and capital stock in the economy for each counterfactual.



Note: The graph presents the housing distribution (left axis) and the corresponding tax rates under *Opt Joint* (right axis) for each housing size. The x-axis reports the housing sizes as multiples of the Baseline median one. The first bin (0 housing value) corresponds to renting. Reported tax rates are percentage shares of housing values. For Baseline the housing tax is flat and equal to 0.62%.

In terms of consumption smoothing, the Optimal Joint tax schedule reduces mean C_{tot} by 10%, but it also reduces the variance by 70%. Importantly it succeeds in reducing both intra-age C_{tot} variance, by 75%, as well as inter-age variance

by 42%. The capacity of the joint tax schedule to smooth consumption over the life-cycle is key in understanding why the welfare gains delivered by the policy are much higher than the *Opt Y* ones, even if under *Opt Y* mean C_{tot} drops by only 9%, and total variance by 57% (table 9).

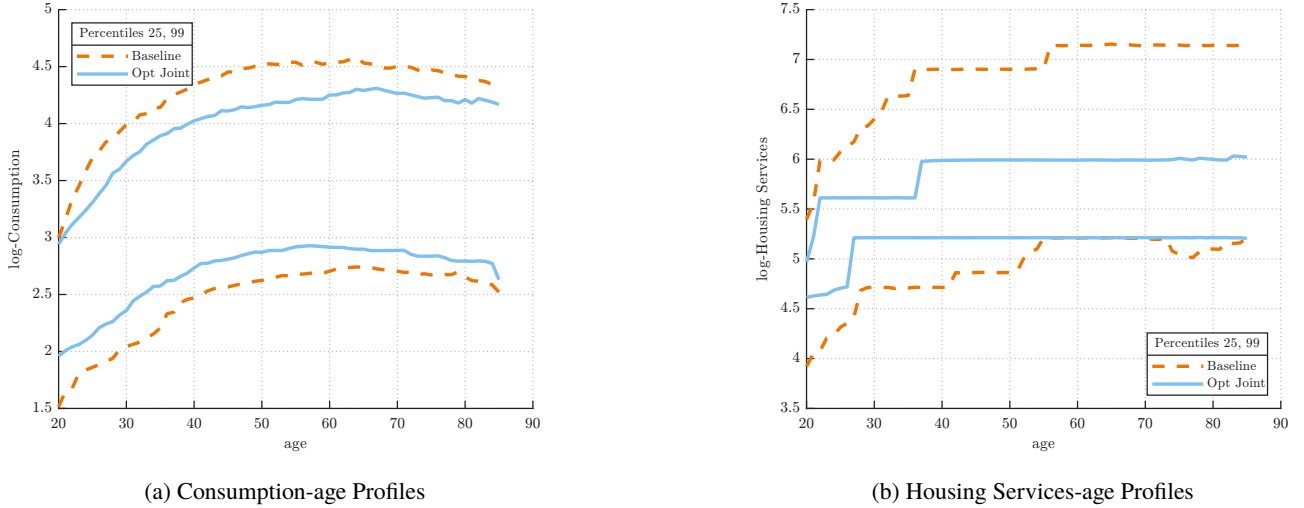
Table 11: C_{tot} Variance Decomposition: Optimal Joint Tax Schedule

	Mean	Total variance	Within-age	Between-age
Baseline	100	100	81	19
<i>Opt Joint</i>	90	31	20	11

Note: Total consumption is defined as $C_{tot} = c^\xi d^{1-\xi}$. The first column reports $E[C_{tot}]$, normalized by its baseline value. The last three columns decompose $Var(\log C_{tot})$ using $Var(\log C_{tot}) = E[Var(\log C_{tot} | j)] + Var(E[\log C_{tot} | j])$ where j denotes age. Total variance and its within- and between-age components are normalized by baseline total variance.

Figure 11 plots the 25th and 99th percentiles of (log) consumption and (log) housing services across age. Looking at the 25th percentile, consumption rises about 30% for those in their 30s, and slightly less than 20% for those over 60s, while housing rises by about 65% for the young. The pattern shifts markedly at the top of the distribution: the 99th percentile of housing services declines sharply—by nearly 67% for older households and 40% for younger ones. Meanwhile, consumption at the 99th percentile declines more modestly, by roughly 25% for young and 20% for old households. Notice that the policy change affects the housing choices at old age both directly, by taxing heavily large houses, and indirectly through general equilibrium effects, as lower house prices reduce the bequest value of housing and therefore the incentive to retain large dwellings into retirement.

Figure 11: Optimal Joint Housing and Income Taxation:
Consumption and Housing Across Age



Note: Housing services (d) are defined as in equation (12). That is: $d = h^r \mathbb{1}_{h=0} + \chi h \mathbb{1}_{h>0}$. The percentiles of consumption and housing services are computed using the invariant distribution for each age value.

Lastly, we explore how tax burdens are allocated across age, income and consumption bins, and compare it with the baseline tax schedule. For each bin, we aggregate taxes paid (or subsidies received) and income received by households, and report the ratio of the two in fig. 12.

In terms of income, the total tax burden as a share of income for households in the upper 20% of the income distribution

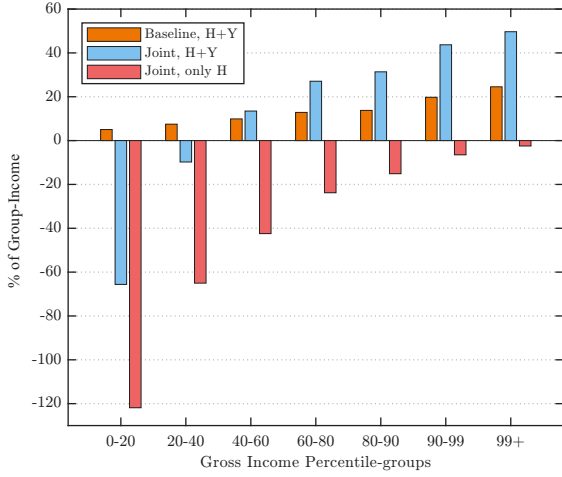
nearly doubles, while the bottom 20% receives massive subsidies equal to 65% of their income (fig. 12a). Crucially, the top 1% faces negligible housing tax, not only because their income is very large (a denominator in the reported ratio), but also because the high progressivity of housing taxation effectively prevents even the richest households from living at the upper h-gridpoints. For households in the bottom 40% of the income distribution housing subsidies are compensating for the high flat income tax, leading to a negative total tax burden. In terms of total consumption (C_{tot}), the picture is rather similar, again with households at the bottom 20% of total consumption receiving net subsidies, and net taxes being higher than baseline at the top C_{tot} bins (fig. 12b).

In fig. 12c we plot the net tax burden as a share of income for different age bins. We find that the policy increases the tax burden for households aged 36-50 and 51-65 by about 15 p.p, due to the higher level of income taxation, but reduces the overall tax burden for retirees (aged 66+), since for the latter age bin, increased housing subsidies offset the effect of rising income taxation.

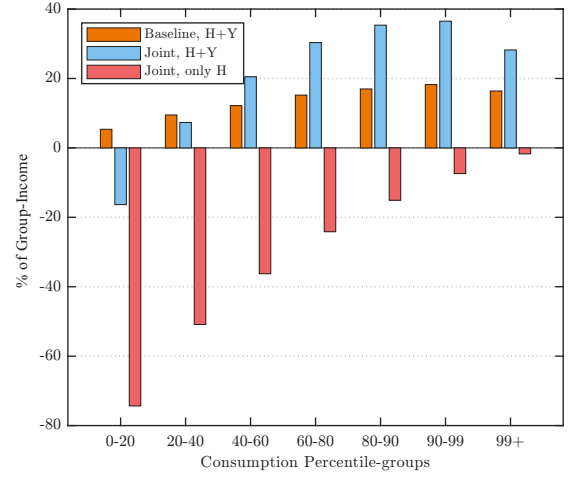
Welfare Gains from the reform are a mix of *Opt Y* and *Opt H* reforms described above. For the newborns with the lowest combination of labor ability z and entrepreneurial ability θ they reach 96.3%, while the worst affected are the ones with the two highest values of z and the highest value of θ who lose 43.1% and 42.6% respectively. Notice that these losses are still below the -55% welfare loss for this group under the *Opt Y* reform. Also in contrast to *Opt Y*, newborns with the middle of labor ability (z_4) and low entrepreneurial abilities face only mild welfare loss, around 3.5%. Overall, a majority of 66% of newborns would be better off if they were born in an economy that had implemented the Opt Joint reform.

Redistribution-Neutral Optimal Taxation: In all the experiments presented so far, welfare gains are large and, as shown, arise to a considerable extent from redistribution. Since the model abstracts from features such as elastic labor supply or human capital accumulation that would limit the desirable degree of overall tax progressivity, we also consider an experiment designed to hold redistribution fixed at its baseline level. Specifically, we jointly optimize the housing and income tax schedules while ruling out subsidies and imposing that the Gini index of total consumption (C_{tot}) equal its baseline value. Welfare gains remain substantial, at 1.98% in consumption-equivalent terms, and the optimal policy again features a much flatter income tax alongside a progressive housing tax. Detailed results are reported in section E.3.

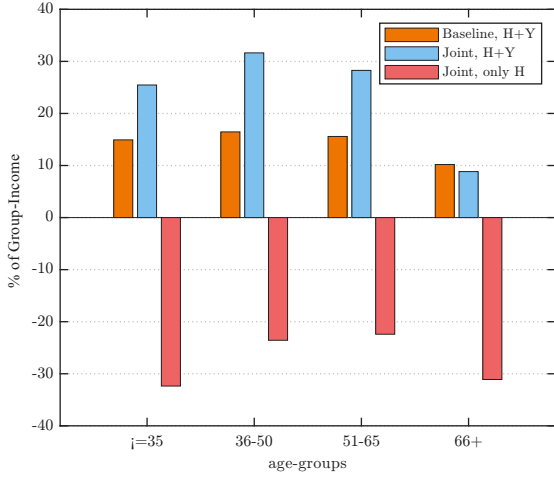
Figure 12: Tax burden and welfare gains under optimal joint taxation



(a) Tax burden by income bins



(b) Tax burden by C_{tot} bins



(c) Tax burden by age bins

	θ_1	θ_2	θ_3	θ_4
z_1	96.3 (5.0)	94.7 (0.6)	86.7 (0.1)	18.6 (0.3)
z_2	48.1 (20.6)	46.7 (2.4)	38.9 (0.2)	-18.1 (1.1)
z_3	17.4 (32.4)	16.6 (3.8)	9.9 (0.4)	-32.9 (1.7)
z_4	-3.5 (20.6)	-4.0 (2.4)	-8.6 (0.2)	-39.7 (1.1)
z_5	-19.9 (5.0)	-20.2 (0.6)	-22.8 (0.1)	-43.1 (0.3)
z_6	-15.4 (0.9)	-15.8 (0.1)	-18.7 (0.0)	-42.6 (0.0)

(d) Newborn welfare gains

Note: Panels (a)–(c) split the population by income, total consumption, or age bins. Each panel aggregates taxes paid and all gross income received by households in each bin, and reports their ratio. For consumption bins, total consumption is defined as $C_{tot} = c^\xi d^{1-\xi}$. 'H+Y' bars denote the total housing and income taxes paid by households, whereas 'only H' bars display the tax burden of the housing tax alone. Panel (d) reports consumption-equivalent welfare gains for newborn households, in percent. Cohort shares are reported in parentheses.

6 Understanding the Mechanisms

6.1 Model Drivers: Entrepreneurship and Homeownership

In section 5 we showed that the optimal joint income and housing tax schedule features a nearly flat income tax and a highly progressive housing tax. Several model mechanisms may drive this result, which we examine in turn.

First, entrepreneurship introduces a trade-off in raising the progressivity of both income and housing taxes. As the tax burden on wealthy households increases, entrepreneurs are discouraged from operating at larger scales, reducing the output of the consumption good and depressing wages. This channel is more pronounced under progressive income taxation. Progressive housing taxation also incentivizes wealthy households to reallocate their portfolios away from

housing and toward financial assets, which partially offsets the adverse effect on the capital stock.

Second, the preference for homeownership implies that there are substantial welfare gains from raising the homeownership rate, even holding the housing stock constant. This can be achieved either by lowering the after-tax cost of housing for poorer households or by increasing their resources through income-tax redistribution. Crucially, owner-occupied housing is an indivisible good that requires a minimum level of investment, which leads households to accumulate financial assets when young in order to eventually become homeowners. The option to take out a mortgage alleviates this need, but only partially: mortgage borrowing carries a higher interest rate wedge and also prevents households from building precautionary savings.

We begin our investigation by removing the option for households to become homeowners. Solving for the jointly optimal income and housing taxes in this restricted model, we find that the optimal mix again combines a near-flat income tax with a highly progressive housing tax, as in the baseline (see table 12). By contrast, when we allow households to become homeowners but remove the option to become entrepreneurs, the optimal tax mix reverses: it features flat housing taxation and a highly progressive income tax—the exact opposite of our baseline results. Therefore, the adverse effects of progressive income taxation on the wealth accumulation of entrepreneurs are the key mechanism that shapes the optimal policy in the model.

Table 12: Optimal Taxation Parameters Sensitivity

	τ_h^l	τ_h^p	τ_y^l	τ_y^p
Baseline (Joint)	1.1	0.15	0.41	0
Only S.E.	1.11	0.12	0.43	0
Only H.O.	0.92	0	0.52	0.87
Baseline, high h_{min}	1.14	0.24	0.25	0.03
Baseline, high χ	1.1	0.22	0.38	0

Note: Tax parameters of the jointly optimal income and housing tax schedules under alternative model assumptions. *Only S.E.* removes homeownership (all households rent); *Only H.O.* removes occupational choice (all households are workers).

We further explore the role of housing by examining, within the baseline model, the sensitivity of results to the homeownership preference parameter and the minimum housing size. Note that a larger minimum size increases the resources households must accumulate to become homeowners. In both experiments, we find that the housing tax schedule becomes more progressive relative to the baseline.

6.2 The Role of the Elasticity of Housing Supply

We now proceed to examine how our results depend quantitatively on the value of housing supply elasticity. In the baseline we rely on the results of [Cavalleri et al. \(2019\)](#) who provide a point estimate of 1.17; nevertheless it is important to know whether the optimality of the progressive housing tax schedule relies on the low value of this parameter. We assume two alternative values for this elasticity: 10 (high supply elasticity scenario) and 0.1 (low supply elasticity scenario).

The elasticity in our model directly pins down the share of land in the construction sector’s production function, as $\epsilon_h = \frac{1-\gamma_L}{\gamma_L}$. Maintaining the normalization $p_h = 1$ at Baseline also requires rescaling the amount of land in the economy $\bar{L}$. Since construction firm’s profits are collected by the government, the government expenditure also needs to be adjusted. The rest of the calibrated parameters are not affected, nor are the simulated moments.

Table 13: Optimal *Joint* Taxation Parameters
Under Different Housing Supply Elasticities

		Baseline Economy	Low elasticity $\epsilon_h = 0.1$	Main estimate $\epsilon_h = 1.17$	High elasticity $\epsilon_h = 10$
<i>Tax Parameters</i>					
τ_h^l	Housing tax level	0.994	1.357	1.103	1.083
τ_h^p	Housing progressivity	0	0.522	0.151	0.1
τ_y^l	Income tax level	0.882	0.423	0.41	0.441
τ_y^p	Income progressivity	0.1	0.001	0	0.001
<i>Tax Structure</i>					
Effective income tax rate (%)		12.8	57.66	59	55.88
Share of hhs paying income tax		72.1	100	100	100
Effective housing tax rate (%)		0.62	-31.55	-11.38	-8.71
Share of hhs paying housing tax		100	0.72	0.09	0.28
<i>Prices</i>					
p_h	Housing price	1	0.19	0.73	0.93
p_r	Rental price	0.05	0.01	0.04	0.06
r	Interest rate (%)	4.09	4.47	5.14	5.28
w	Wage	1	0.98	0.95	0.95
<i>Aggregate Variables</i>					
Production		1	1	0.96	0.93
Capital		1	0.94	0.85	0.83
Housing stock		1	0.85	0.75	0.76
Rental stock		1	0.09	0.29	0.47
Net wealth		1	0.47	0.67	0.76
Share of homeowners		72.4	97.5	90.8	85.8
Welfare ($\Delta\%$)		–	38.44	20.43	14.31

The optimal *Joint* counterfactuals for the two alternative scenarios are reported in table 13. The results show that the housing supply elasticity has a sizable quantitative effect on the degree of housing-tax progressivity and on the associated welfare gains. Nevertheless, in both cases the optimal income tax remains essentially flat, and redistribution is carried out primarily through housing subsidies.

Welfare gains are substantially larger when housing supply is less elastic, reaching 38.5% when $\epsilon_h = 0.1$. In this case, housing supply is close to fixed, so progressive housing taxation generates large redistributive gains. By reducing the demand for high-value properties, the policy induces a sharp decline in house prices, which fall by roughly 81%. This makes housing considerably more affordable for lower-income households and leads to a very high homeownership rate, equal to 97.5%. The reallocation is also reflected in the distribution of housing demand: most households choose house sizes above the baseline mean, while there is no demand for the four largest housing grid points.

Even in an environment with very elastic housing supply, the *Joint* counterfactual delivers sizable welfare gains, equal to about 14%. In this case, the decline in house prices is much smaller, around 7%, and is driven mainly by the reduction in aggregate housing demand. Hence, the housing supply elasticity is important for determining the magnitude of the welfare gains, but it is not the main driver of the qualitative results. The direction of the relevant mechanisms remains the same across different values of the housing supply elasticity: optimal redistribution continues to rely on a

progressive housing tax schedule combined with an essentially flat income tax.

7 Conclusion

We examine the impact of introducing a progressive housing tax in a heterogeneous-agent model that incorporates housing and entrepreneurship. Using Spain as our laboratory, our findings suggest that the optimal progressive housing tax schedule enhances the welfare of newborns by 10% in consumption-equivalent terms—slightly below the gains from optimizing the income tax schedule (11.9%). While progressive income taxation reduces consumption variance within age cohorts at the expense of a diminished capital stock, progressive housing taxation fosters capital deepening and shifts consumption from wealthy, older households toward poorer, younger ones.

When jointly optimizing the housing and income tax schedules, we find that the optimal policy involves imposing a flat, high income tax rate and redistributing the resulting revenues through a highly progressive housing tax. This combined approach yields welfare gains of 20.4%, primarily driven by improved consumption smoothing both across the life cycle and within age groups. Moreover, joint optimization mitigates the decline in capital stock typically associated with high income tax rates. These optimality results critically depend on the inclusion of entrepreneurship in the model: income tax-based redistribution imposes efficiency costs by discouraging business expansion among entrepreneurs. In contrast, progressive housing taxation incentivizes the wealthiest households to increase deposit savings, thereby promoting capital deepening and reducing house prices—benefiting younger and poorer households. The more inelastic the housing supply, the stronger the equilibrium effects on prices and the greater the resulting welfare gains.

Our findings broadly align with and extend the existing literature on housing taxation. As shown in [Balke et al. \(2025\)](#) and [Rotberg \(2022\)](#), housing taxes promote capital deepening and reallocate consumption from older to younger households. In our framework, the costs of progressive income taxation stem from the inclusion of entrepreneurship, but other mechanisms – such as endogenous labor supply or human capital accumulation— would likely reinforce our results. Indeed, [da Costa and Santos \(2023\)](#) demonstrate similar dynamics in the context of progressive consumption taxation within an overlapping generations Bewley model featuring endogenous labor supply. In this sense, housing taxation in our model can also be interpreted as a form of consumption tax.

Our analysis highlights the potential of progressive housing taxation to generate substantial welfare gains. However, these results warrant cautious interpretation. First, a key mechanism in the model is that reducing housing demand among wealthy households lowers house and rental prices for poorer households. This link may weaken if rich and poor households occupy distinct housing segments (e.g., holiday homes vs primary residences). In such cases, progressive housing taxation may still yield benefits through portfolio reallocation by the rich, but the welfare gains for poorer households could be attenuated. Second, our model abstracts from house price uncertainty. High transaction costs combined with price volatility may temper the welfare benefits of the policy. Third, we treat savings in rental housing as equivalent to deposits. Introducing frictions such as illiquidity or minimum investment thresholds for real estate could alter the results. On one hand, higher returns on real estate investments may amplify the redistributive potential of progressive taxation. On the other hand, illiquidity could reduce the optimal degree of housing tax progressivity. These are promising avenues for future research.

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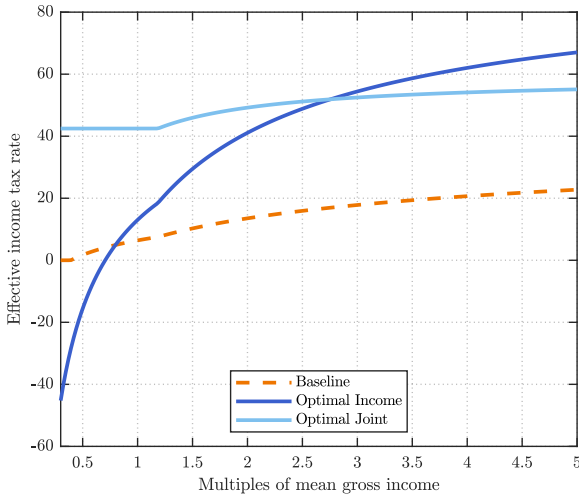
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A Additional figures

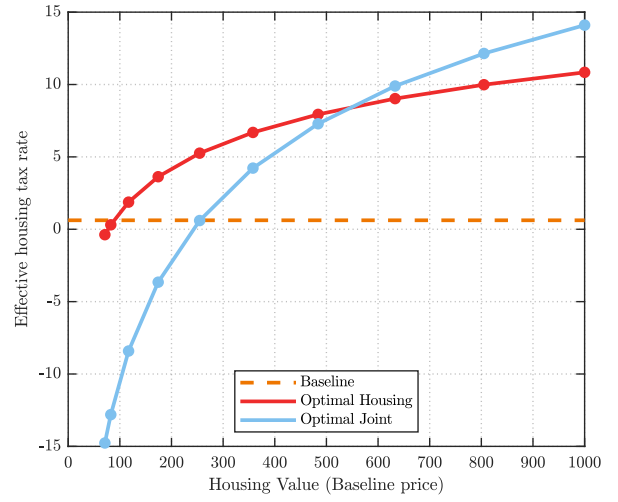
Table A1: Housing portfolio share by Labor (z) and Entrepreneurial Ability (θ) Rank

(a) Baseline					(b) Optimal Income Tax ($Opt Y$)					(c) Optimal Housing Tax ($Opt H$)				
	θ_1	θ_2	θ_3	θ_4		θ_1	θ_2	θ_3	θ_4		θ_1	θ_2	θ_3	θ_4
z_1	53.7	51.2	40.3	29.8	z_1	79.4	76.4	67.3	58.1	z_1	53.3	47.3	28.8	16.4
z_2	60.9	57.9	44.5	29.6	z_2	74.6	72.9	67.1	56.6	z_2	62.0	56.2	33.9	16.0
z_3	60.9	58.2	46.2	29.0	z_3	63.4	63.0	60.1	52.9	z_3	51.1	48.1	33.8	15.3
z_4	52.9	51.7	44.1	29.3	z_4	51.7	51.5	50.3	47.8	z_4	42.2	40.6	30.9	15.4
z_5	40.8	40.5	37.7	29.4	z_5	41.8	41.8	41.4	39.2	z_5	24.8	24.5	21.9	14.1
z_6	31.5	31.4	30.6	26.4	z_6	33.6	33.6	33.6	33.5	z_6	15.9	15.8	15.2	12.3

Figure A1: Optimal Tax Functions. Comparison with the optimal joint tax schedule.



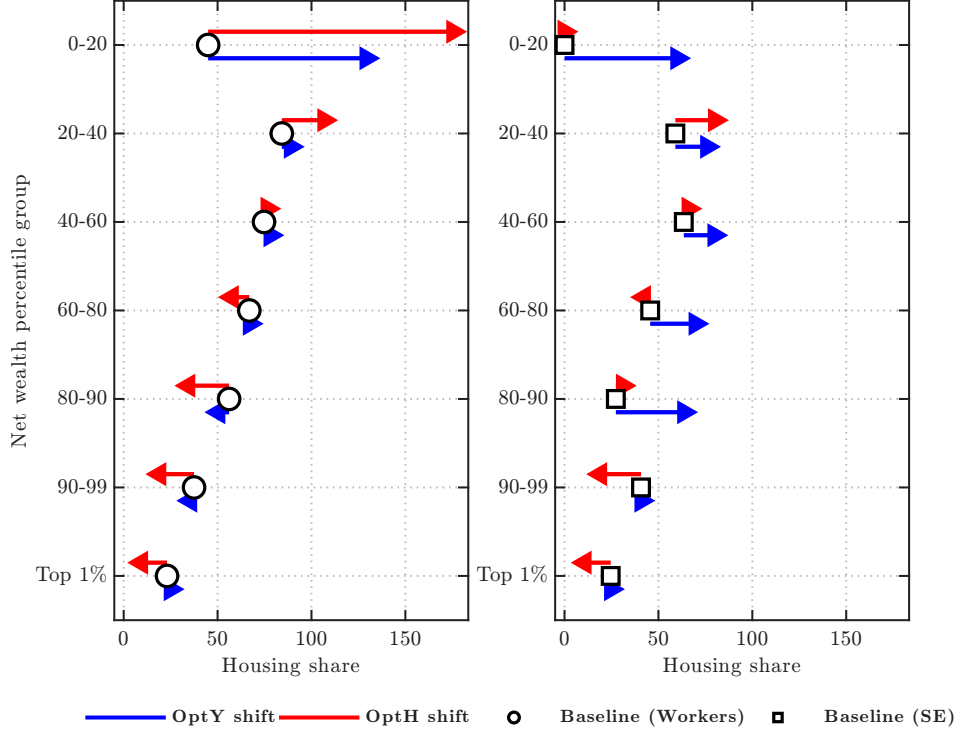
(a) Optimal Income Tax Function



(b) Optimal Housing Tax Function

Note: this figure plots the optimal tax functions described in section 5. Panel (a) plots the optimal income tax function ($Opt Y$ in the main text), the optimal income schedule under the joint optimization ($Opt Joint$), and the baseline income tax function. Panel (b) describes the housing tax schedules that correspond to the same two counterfactuals. The circles denote tax rates at the housing grid points.

Figure A2: Portfolio shares by net worth bins



Note: the figure depicts portfolio shares for workers (Left Panel) and entrepreneurs (Right Panel) across the wealth distribution and counterfactuals.

B Further model detail

B.1 Dynamic problem of retired households

The problem of the retirees is divided in two stages similarly to the working-age maximization problem described in Section 2. In the first stage, a retired household decides on the amount of current non-durable consumption c , the savings for the next stage s , and current size of their dwelling if the household is a renter. That is, if the current housing holdings are zero ($h = 0$).

$$\begin{aligned}
 \mathcal{V}_j(a, h, z, \theta) &= \max_{c, s, h^r} u_j(c, d) + \tilde{\mathcal{V}}_j(s, h, z, \theta) \\
 \text{s.t. } j &\geq R \\
 c(1 + \tau_c) + s + (p_r h^r + t_h(h^r) p_h) \mathbb{1}_{h=0} + m &\leq a + r(1 - \tau_k) a \mathbb{1}_{\{a \geq 0\}} + y_p - p_h h \delta_h - p_h t_h(h) \\
 y_p &= w \bar{p} q(R - 1) z - t_y(w \bar{p} q(R - 1) z) \\
 d &= h^r \mathbb{1}_{h=0} + \chi h \mathbb{1}_{h>0} \\
 s &\geq 0, \quad h^r \geq 0
 \end{aligned} \tag{30}$$

The main difference with respect to the first stage of working-age households resides in the lack of occupational choice. Instead, retirees receive a fixed pension, which is also subject to progressive income taxation. A household may enter the very first retirement period with some debt $a < 0$, in which case it must be fully repaid within the period.

The second stage also resembles the one of working-age households. Nevertheless, notice that retired households are no longer subject to labor productivity or entrepreneurial ability shocks. Thus, the only uncertainty the household faces is about survival for the next period. Additionally, we restrict retirees from taking mortgages in order to finance the purchase of a dwelling.

$$\begin{aligned}
\tilde{\mathcal{V}}_j(s, h, z, \theta) &= \max_{h', a'} \beta \zeta_j \mathcal{V}_{j+1}(a' + B_{j+1}(z), h', z, \theta) + \beta (1 - \zeta_j) \Phi(b) \\
\text{s.t. } j &\geq R \\
p_h h' + p_h C_{adj}(h, h') + a' &\leq s + p_h h \\
C_{adj}(h, h') &= \mathbb{1}_{h \neq h'} (\psi_s h + (\psi_b + \tau_h^t) h') \\
b &= p_h h' + a' - \psi_s p_h h' - t_{inh}(p_h h' + a') \\
a' &\geq 0 \quad , \quad h' \in \hat{\mathcal{H}} = \{0, h_1, h_2, \dots, h_N\}
\end{aligned} \tag{31}$$

C Computation

C.1 Model Computation

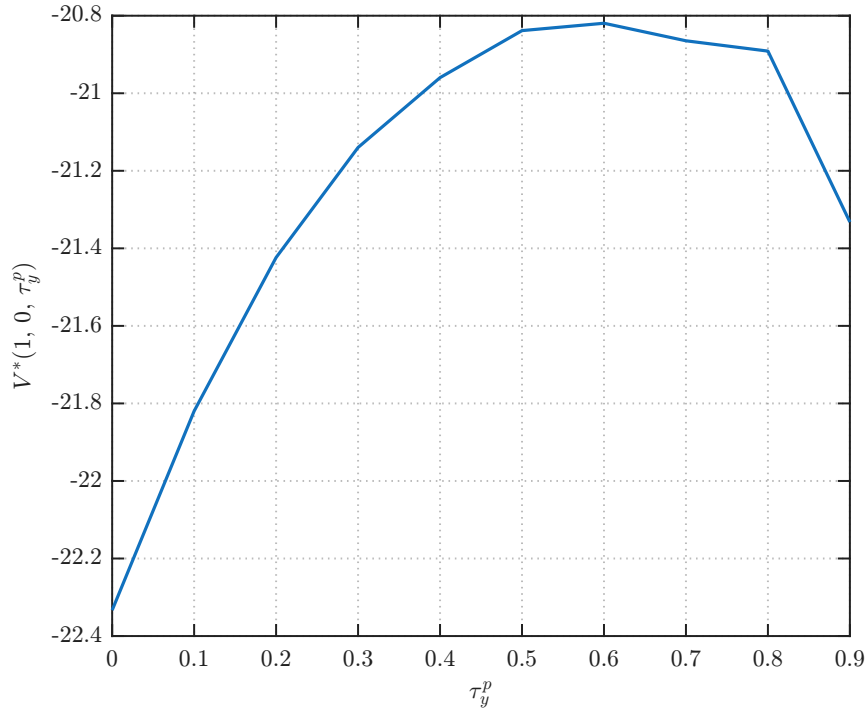
To solve the model we rely on the Markov chain approximation method developed in [Bakota and Kredler \(2022\)](#), which we adapt to deal with features specific to our environment. The method leverages the insight that during a single period most of the agents move no further away than a single gridpoint on the financial asset grid, allowing the use of first-order conditions in those cases. In infinite-horizon settings, reducing the model period can ensure that it is never optimal to move beyond one gridpoint. However, in finite-horizon problems, wealthy households approaching the end of life have strong incentives to deplete their assets. To deal with this behavior, we extend the original algorithm by searching across segments of the financial asset grid for state values where agents may optimally move more than one gridpoint. Because the value function features kinks—arising from occupational choices and the discrete nature of housing—this extension requires searching over non-adjacent grid segments whenever such discontinuities are present. To limit the computational burden, we restrict the extended search to two cases: for kinks above the current gridpoint, we search only if the move is affordable; for kinks below, we exploit the monotonicity of the policy function and search only if the optimal saving choice at the previous gridpoint lies below the kink.

Another computationally intensive component of the paper is the joint maximization problem. Since we focus on budget-neutral reforms, we search over the three-dimensional space spanned by $(\tau_h^l, \tau_h^p, \tau_y^p)$ and for each combination of them we find the implied τ_y^l that balances the government budget. This is a non-differentiable and potentially non-convex problem, with a costly to evaluate objective function. To address this, we employ global optimization algorithms that require only continuity of the objective function. Specifically, we use the Dividing Rectangles (DIRECT) algorithm as described in [Jones and Martins \(2021\)](#) which guarantees convergence to the global optimum as the number of function evaluations increases. We stop the algorithm when the maximum point has not changed after 800 function evaluations.

To build intuition for the problem, it is instructive to examine the shape of the objective function, taking the *Opt Joint* experiment as an illustrative example. Figure C1 plots the expected value function of a newborn for $\tau_h^p = 0$, $\tau_h^l = 1$, and varying τ_y^p ; the resulting function is concave with a peak around $\tau_y^p = 0.6$. Figure C2b plots the optimal τ_y^p for different values of τ_h^p and τ_h^l , with the red box indicating the region over which the DIRECT algorithm is applied.¹⁸ The figure reveals that the optimal τ_y^p is inversely related to τ_h^p . Moreover, for very high levels of housing subsidies—whether driven by high τ_h^l or high τ_h^p —the budget does not clear. At the boundary of the feasible region, the optimal τ_y^p is close to zero: since subsidies are permitted, a flat income tax is more efficient for revenue collection than a progressive one, and large housing subsidies can only be financed through sufficiently high flat income taxes. Note also that allowing for regressive income taxation ($\tau_y^p < 0$) would likely expand the set of fiscally-feasible allocations, as it would incentivize entrepreneurs to produce more, generating additional housing tax revenues in the process. Figure C2a plots the maximum value of the objective function for given τ_h^p and τ_h^l . The surface is unimodal and hill-shaped: taken together, the two figures indicate that high values of the objective function are attainable under both high τ_h^p and high τ_y^p , but the global peak occurs at $\tau_y^p = 0$.

¹⁸Outside the box, graphs are plotted using sparse grids; inside the box, the grid is enriched by the points evaluated during the DIRECT algorithm.

Figure C1: A slice of the obj. function for $\tau_h^p = 0, \tau_h^l = 1$ (Opt Joint)

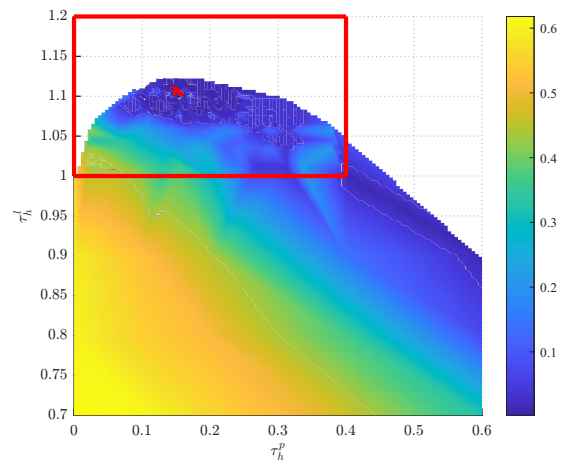
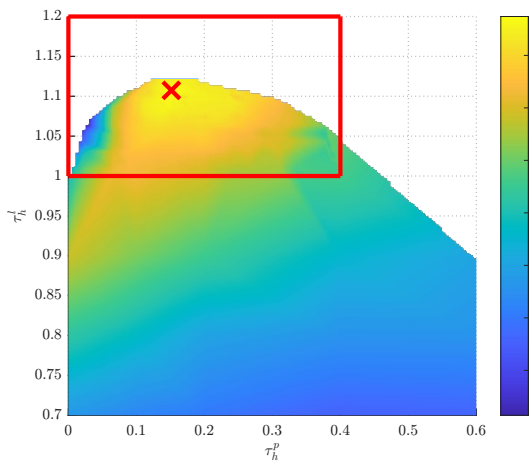


Note: The figure plots the expected utility of a newborn as a function of income tax progressivity τ_y^p , holding housing tax parameters fixed at $\tau_h^p = 0$ and $\tau_h^l = 1$.

Figure C2: Shape of the objective function (Opt Joint)

(a) maximum Exp. Value of Newborn given τ_h^p, τ_h^l

(b) Optimal τ_y^p given τ_h^p, τ_h^l



Note: The left panel plots the maximum attainable expected utility of a newborn over the space of housing tax parameters (τ_h^p, τ_h^l) , with the red marker indicating the global optimum. The right panel plots the corresponding optimal income tax progressivity τ_y^p . The red box delimits the region over which the DIRECT algorithm is applied; outside this region, values are computed on sparse grids, using 1-dimensional golden section search for τ_y^p .

D Calibration Details

D.1 Regression results

In Table D1 we report the linear regression coefficients that we used in Section 3 to construct a smoother function for each series. We only regress on age because we are only interested in obtaining the mean outcome by age. The first regression result corresponds to the death probability. The data is taken directly from the INE and expressed in percentages.

The variable s_j^{inh} stands for the inheritance share by age, measured in percentage points. For the largest assets (such as housing or a business) the EFF reports whether those assets were inherited or received as gifts. Nevertheless we cannot directly use this variable since we would overestimate the share of inheritances in later life stages. Thus, we construct an indicator variable that tells us if a household received a significant inheritance (such as a house) in the last 3 years. Pooling observations across survey waves and age groups, we fit a fourth-degree polynomial to recover a smooth age profile, which delivers the double-humped shape observed in the data.

The third column displays the coefficients for the equivalence scale. We define the equivalence scale using the standard OECD definition.

Lastly, we regress labor income onto a second-degree age polynomial. We define labor income as wages, in-kind benefits as well as unemployment benefits. We use these results to retrieve the age-specific component of labor productivity, common to all households.

Table D1: Regression Results

	$\ln(1 - \zeta_j)$	s_j^{inh}	$\ln(e_j)$	$\ln(y)$
c	-2.3063	26.259	-0.6015	9.0037
age	-0.1962	-2.3549	0.0615	0.0537
age ²	0.0071	0.0779	-0.001	-0.0006
age ³	0.00008	-0.0010	0.000004	
age ⁴	0.0000003	0.000004		
N	76	527	6,355	3,305
R ²	0.999	0.1724	0.1304	0.0065

D.2 Calibrated parameters

The full set of labor ability levels z and the associated transition matrix is given by:

$$z = \{z_1, \dots, z_6\} = \{0.2475, 0.4975, 1, 2.01, 4.04, 9.7775\}$$

$$\Lambda_z(z' | z) = \begin{pmatrix} 0.9204 & 0.0751 & 0.0023 & 0.0000 & 0.0000 & 0.0021 \\ 0.0188 & 0.9216 & 0.0564 & 0.0012 & 0.0000 & 0.0021 \\ 0.0004 & 0.0376 & 0.9219 & 0.0376 & 0.0004 & 0.0021 \\ 0.0000 & 0.0012 & 0.0564 & 0.9216 & 0.0188 & 0.0021 \\ 0.0000 & 0.0000 & 0.0023 & 0.0751 & 0.9204 & 0.0021 \\ 0 & 0 & 0.1989 & 0 & 0 & 0.8011 \end{pmatrix}$$

The inheritance statutory tax schedule is taken from [Fuster \(2022\)](#) and reproduced in table [D2](#) below.

Table D2: Inheritances: statutory tax schedule

	Bequest Brackets (thousand Euros)							
	0–8	8–16	16–24	24–32	32–40	40–48	48–56	56–64
Tax Brackets	0–8	8–16	16–24	24–32	32–40	40–48	48–56	56–64
Tax Rates	0.0765	0.0850	0.0935	0.1020	0.1105	0.1190	0.1275	0.1360
Tax Brackets	64–72	72–80	80–130	130–160	160–240	240–400	400–800	+800
Tax Rates	0.1445	0.1530	0.1615	0.1870	0.2125	0.2550	0.2975	0.3400

E Additional Results

E.1 No subsidies case

In section 5 we have presented three experiments for the optimal housing tax schedule (*Opt H*), the optimal income tax schedule (*Opt Y*) and the optimal *Joint* counterfactual which delivers the optimal combination of both tax schedules. In all of those cases we have allowed for potential subsidies in both the income and housing tax functions. Here we restrict the progressive schedules so as to not allow subsidies.

Without subsidies housing progressivity has a larger role in the model, relative to income progressivity. As shown in table E1 the welfare gains from the *Opt H* scenario (6.2%) are much higher than that from *Opt Y* (0.41%). The reason is that in the baseline the income-poorest households do not pay income taxes, so higher progressivity is not relieving the tax burden of that 28% of the households that were not paying taxes in the baseline. Housing taxation, on the contrary, improves the welfare of the poorest households directly *via* lower tax burden and indirectly through general equilibrium effects: lower housing price and interest rate paired with higher wages.

Table E1: Housing vs Income Taxation Results – Without Subsidies

		Baseline	Optimal Counterfactual		
			Income	Housing	Joint
<i>Tax Parameters</i>					
τ_h^l	Housing tax level	0.9938	0.9938	0.9661	0.9741
τ_h^p	Housing progressivity	0	0	0.0835	0.0611
τ_y^l	Income tax level	0.8817	0.8869	0.8817	0.8965
τ_y^p	Income progressivity	0.0996	0.1589	0.0996	0.1222
<i>Tax Structure</i>					
Effective income tax rate (%)		12.8	9.632	9.789	9.017
Share of hhs paying income Tax		72.16	45.124	72.002	67.386
Effective housing tax rate (%)		0.620	0.620	2.544	2.442
Share of hhs paying housing Tax		100	100	63.840	67.476
<i>Prices</i>					
p_h	Housing price	1	0.98	0.76	0.80
p_r	Rental price	0.05	0.05	0.029	0.032
r	Interest rate (%)	4.09	4.22	3.16	3.29
w	Wage	1	0.99	1.05	1.04
<i>Aggregate Variables</i>					
Production		1	0.99	1.06	1.05
Aggregate capital		1	0.96	1.02	1.00
Real housing stock		1	0.99	0.67	0.72
Rental stock		1	0.90	0.60	0.65
Net Wealth		1	0.97	0.77	0.80
Share of homeowners		72.40	73.90	79.00	77.50
Welfare ($\Delta\%$)		–	0.41	6.20	6.35

To clarify the distributional effects of these two policies, table E2 reports welfare changes for newborns by initial pro-

ductivity and entrepreneurial shock.¹⁹ Under the *Opt H* scenario, households with the lowest initial abilities experience the largest welfare gains, while the highest ability groups suffer significant welfare losses. In contrast, *Opt Y* reduces welfare for both high- and low-ability households. Low-ability households do not benefit from the policy because they: 1) gain little, or nothing, from tax reductions; 2) face lower wages; and 3) are burdened by higher interest rates that increase rental and mortgage costs.

Table E2: Newborns Welfare Gains by Labor & Entrepreneurial Ability Rank
(a) Optimal income tax (*Opt Y*)

	θ_1	θ_2	θ_3	θ_4
z_1	-0.4 (5.0)	-0.4 (0.6)	-0.4 (0.1)	-2.8 (0.3)
z_2	0.1 (20.6)	0.1 (2.4)	0.3 (0.2)	-3.6 (1.1)
z_3	1.2 (32.4)	1.3 (3.8)	1.1 (0.4)	-4.7 (1.7)
z_4	0.9 (20.6)	0.9 (2.4)	0.2 (0.2)	-6.3 (1.1)
z_5	-1.1 (5.0)	-1.1 (0.6)	-1.7 (0.1)	-7.7 (0.3)
z_6	-2.5 (0.9)	-2.5 (0.1)	-2.9 (0.0)	-8.3 (0.0)

(b) Optimal housing tax (*Opt H*)

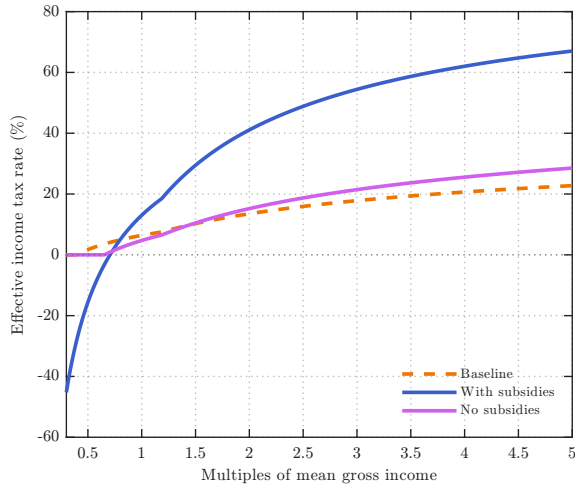
	θ_1	θ_2	θ_3	θ_4
z_1	17.7 (5.0)	17.0 (0.6)	14.5 (0.1)	-4.4 (0.3)
z_2	13.3 (20.6)	12.5 (2.4)	9.6 (0.2)	-10.7 (1.1)
z_3	6.6 (32.4)	6.0 (3.8)	3.1 (0.4)	-15.2 (1.7)
z_4	-0.6 (20.6)	-0.9 (2.4)	-3.7 (0.2)	-19.0 (1.1)
z_5	-9.7 (5.0)	-9.9 (0.6)	-12.0 (0.1)	-22.0 (0.3)
z_6	-13.1 (0.9)	-13.2 (0.1)	-14.1 (0.0)	-23.5 (0.0)

Note: The cohort share of households for each combination of the labor z and entrepreneurial ability θ shock is reported in parentheses.

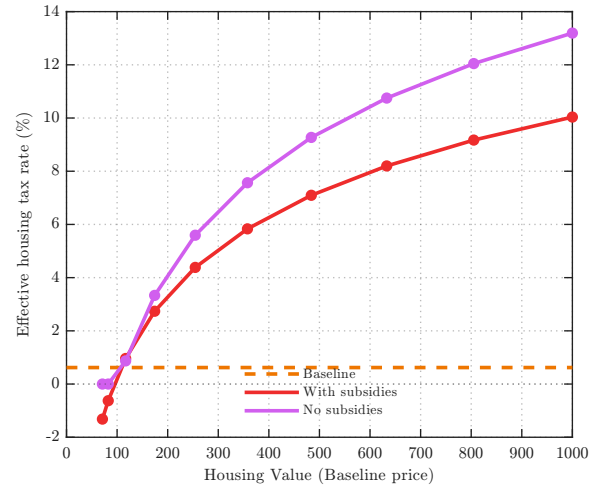
The biggest difference between the counterfactuals with and without subsidies is in the results for the *Joint* taxation experiment. In the case with subsidies, the optimal tax schedule was to impose a high and flat income tax rate while using a progressive housing tax to redistribute subsidies towards housing. Without subsidies the optimal schedule relies on a higher progressivity in both income and housing taxation. The reason is that without subsidies the housing taxation cannot be used to re-distribute resources directly. Thus, there is no need for a pool of fiscal resources generated by income taxes to later re-distribute *via* housing subsidies.

Nevertheless, the counterfactuals without subsidies still point to a potential role for progressivity in housing taxation. First, it generates significantly higher welfare gains than its *Opt Y* counterpart. This reveals that there is little room for welfare gains from more progressive income taxation without direct transfers, which is not the case with housing taxation. Second, the *Joint* counterfactual is basically equivalent to *Opt H* in its welfare gains and overall aggregate effects, with a slightly higher income progressivity. But the main channel is still a higher capital intensity that comes from redistribution from housing towards the financial assets. This leads to beneficial general equilibrium effects due to lower housing prices (-20%), higher wages (+4%) and lower interest rates (-0.8 p.p.).

¹⁹Because these shocks are persistent, they are good predictors of households' relative income and wealth over the life cycle.



(a) Optimal income tax schedules (*Opt Y*)



(b) Optimal housing tax schedules (*Opt H*)

Figure E1: Optimal tax schedules with and without subsidies

E.2 Optimal Flat Housing, Progressive Income Tax

In most economies, housing taxation tends to be nearly flat, whereas income taxation is typically highly progressive. However, as demonstrated in section 5.2, the optimal tax configuration in our model reverses this pattern—favoring a flat income tax and a progressive housing tax. This raises a natural question: how closely can economies approximate the optimal outcome while maintaining a flat housing tax?

To explore this, we compute the optimal budget-neutral reform under a regime featuring a flat housing tax combined with a progressive income tax (*Flat H, Prog. Y*). The results, presented in table E3, are compared against the jointly optimal reform. The flat housing tax reform entails a substantial tax rate of 7.28% of property value, though its tax burden is mitigated by a 34% decline in house prices. Meanwhile, the income tax schedule becomes markedly progressive, with $\tau_y^p = 0.62$.

This configuration yields a welfare gain of 15.36% in consumption-equivalent terms—approximately 5.1 percentage points below the fully optimal policy, yet about 5.4 percentage points higher than the housing-only reform. Aggregate outcomes broadly mirror those under the joint optimum, though capital stock and output respond more strongly. The sharper decline in aggregate capital, and thus production, is primarily driven by a two-thirds decline of capital demand among self-employed agents (see fig. E2b).

The housing distribution among owners becomes more dispersed: roughly 12% of households occupy the smallest allowable owner-occupied units (slightly above the baseline), and a modest uptick is observed in the share of households residing in mid-to-large housing grid points (fig. E2a). Interestingly, homeownership is higher than under *Opt Joint*, reflecting the potential of income-tax redistribution to raise homeownership. By contrast, *Flat H* increases significantly the effective rental cost, which if we include the flat housing tax is around 0.09 while it is 0.056 in the baseline.

Looking at housing services over age (fig. E3b), we find the 99th percentile under *Flat H, Prog Y*, closely tracks the 99th percentile of *Joint*. Poor and young households though fare significantly worse than under *Joint*, with the p25 for households in their 30s of housing services being 40% lower, i.e. at same level as baseline. For older households, the p25 of housing services is unchanged across the 3 tax schedules.

The picture is different when looking at the consumption over age of rich and poor households (fig. E3a). Poor house-

Table E3: Flat Housing, Progressive Income tax schedule

	Baseline	Flat H, Prog. Y	Joint
<i>Tax Parameters</i>			
τ_h^l Housing tax level	0.994	0.927	1.103
τ_h^p Housing progressivity	0	0	0.151
τ_y^l Income tax level	0.882	0.803	0.41
τ_y^p Income progressivity	0.1	0.622	0
<i>Tax Structure</i>			
Effective income tax rate (%)	12.8	2.35	59
Share of hhs paying income tax	72.16	32.678	100
Effective housing tax rate (%)	0.620	7.284	-11.377
Share of hhs paying housing tax	100	100	0.093
<i>Prices</i>			
p_h Housing price	1	0.66	0.73
p_r Rental price	0.050	0.042	0.042
r Interest rate (%)	4.09	5.72	5.14
w Wage	1	0.929	0.952
<i>Aggregate Variables</i>			
Production	1	0.92	0.96
Aggregate capital	1	0.63	0.85
Real housing stock	1	0.69	0.75
Rental stock	1	0.09	0.29
Net Wealth	1	0.58	0.67
Share of homeowners	72.40	95.50	90.80
Welfare ($\Delta\%$)	–	15.36	20.43

holds consume slightly more under the *Flat H, Prog Y* tax schedule than under the optimal Joint; since they delay moving to a larger house, they spend more on the consumption good. In contrast, rich households already by age of 30 consume 30% less than they would under the *Joint* policy, and the gap remains at roughly 30% during the rest of the life-cycle.

In contrast, the *Opt Joint* policy delivers a more targeted intervention: it achieves the same reduction in housing demand among wealthy households, but with a substantially smaller impact on their consumption, as detailed in section 5.2.

Figure E2: Flat Housing, Progressive Income tax schedule:
Housing Distribution and Resources Allocation

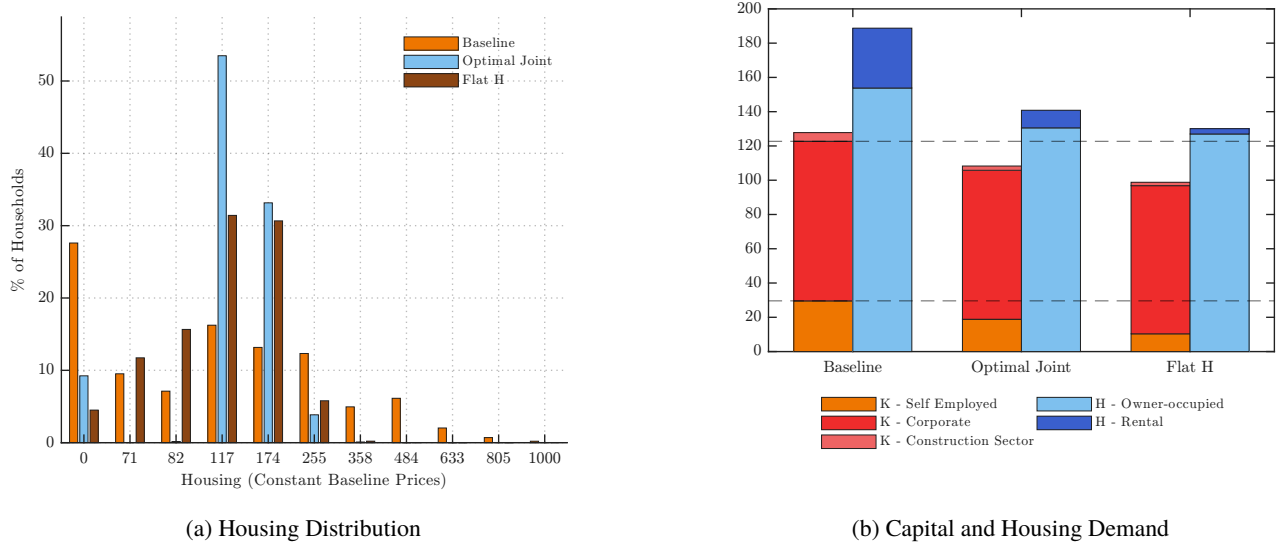
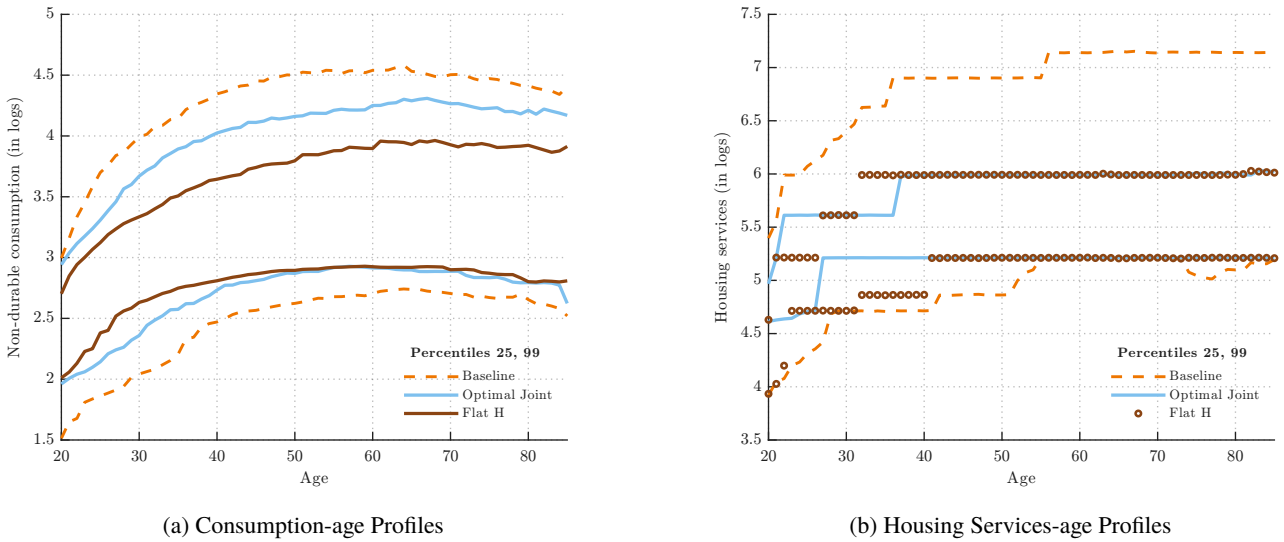


Figure E3: Flat Housing, Progressive Income tax schedule:
Consumption and Housing Across Age



Note: Housing services (d) are defined as in equation (12). That is: $d = h^r \mathbb{1}_{h=0} + \chi h \mathbb{1}_{h>0}$. The percentiles of consumption and housing services are computed using the invariant distribution for each age value.

E.3 Redistribution-Neutral Optimal Taxation

The baseline results feature a highly redistributive tax policy. However, the model abstracts from several features that would reduce the optimal degree of redistribution, such as elastic labor supply or human capital accumulation. To isolate the efficiency implications of the tax reform from its redistributive ones, we impose an additional constraint: we search for the jointly optimal housing and income tax schedules, ruling out subsidies, and requiring that the Gini index of total consumption ($C_{tot} = c^\xi d^{1-\xi}$) match its baseline value. Staying close to actual Spanish tax design in this way, we find that the optimal policy involves a considerably flatter income tax (τ_y^p falling from 0.1 to 0.037), now paid by all households, alongside a higher and more progressive housing tax (see table E4). Housing tax rates range

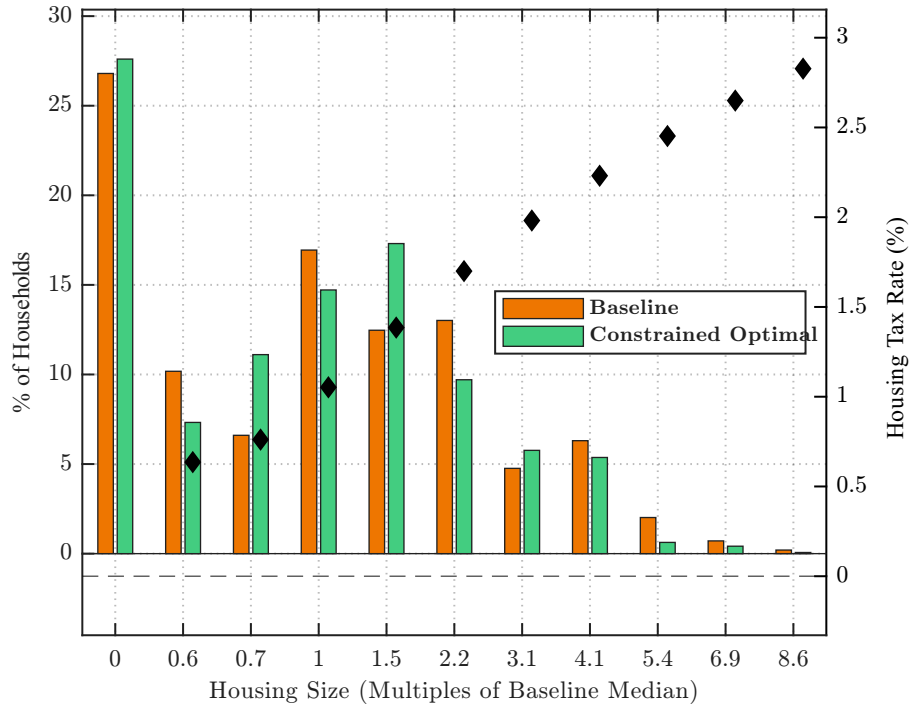
from 0.6% for small houses to 2.8% for the very large ones. Notably, even under this constraint, welfare gains remain substantial at 1.98% in consumption-equivalent terms. These gains do not stem from increased homeownership—which is stable—but from higher production of the consumption good (up 3.5%) and higher wages (up 3.2%). The result that the income-tax is no longer flat is in accordance with the unconstrained joint optimization of the housing and income tax when subsidies are not allowed (see section E.1).

Table E4: Redistribution-Neutral Optimal Taxation

		Baseline	Optimal Gini C_{tot} constrained
<i>Tax Parameters</i>			
τ_y^l	Income Tax Level	0.882	0.905
τ_y^p	Income Tax Prog	0.1	0.037
τ_h^l	H Tax Level	0.994	0.989
τ_h^p	H Tax Prog	0	0.008
<i>Tax Structure</i>			
Effective income tax rate (%)		12.803	9.943
Share of hhs paying income tax		72.161	100
Effective housing tax rate (%)		0.620	1.391
Share of hhs paying housing tax		100	98.616
Gini of consumption		0.385	0.385
<i>Prices</i>			
p_h	Housing price	1	0.973
p_r	Rental price	0.05	0.042
r	Interest rate (%)	4.09	3.475
w	Wage	1	1.032
<i>Aggregate Variables</i>			
Production		1	1.04
Financial wealth		1	1.08
Capital		1	1.11
Housing stock		1	0.92
Rental stock		1	0.97
Share of homeowners		72.40	72.40
Welfare ($\Delta\%$)		–	1.976

* Total consumption is defined as $C_{tot} = c^\xi d^{1-\xi}$

Figure E4: Housing distribution under Redistribution-Neutral Optimal Taxation



Note: The graph presents the housing distribution (left axis) and the corresponding tax rates (right axis) under the Redistribution-Neutral Optimal Tax-Schedule for each housing size. The x-axis reports the housing sizes as multiples of the Baseline median one. The first bin (0 housing value) corresponds to renting. Reported tax rates are percentage shares of housing values. For Baseline the housing tax is flat and equal to 0.62%.